

Estimation of Optical Turbulence (C_n^2) and Seeing from MM5 data in Paranal and Armazones (E-ELT)

10 November to 24 of December of 2007

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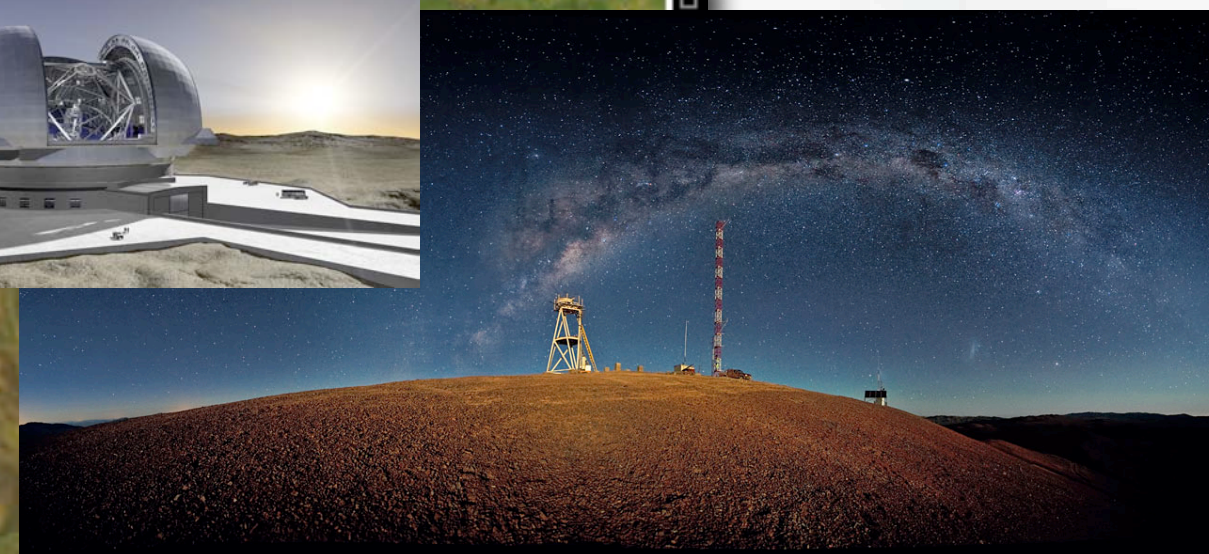
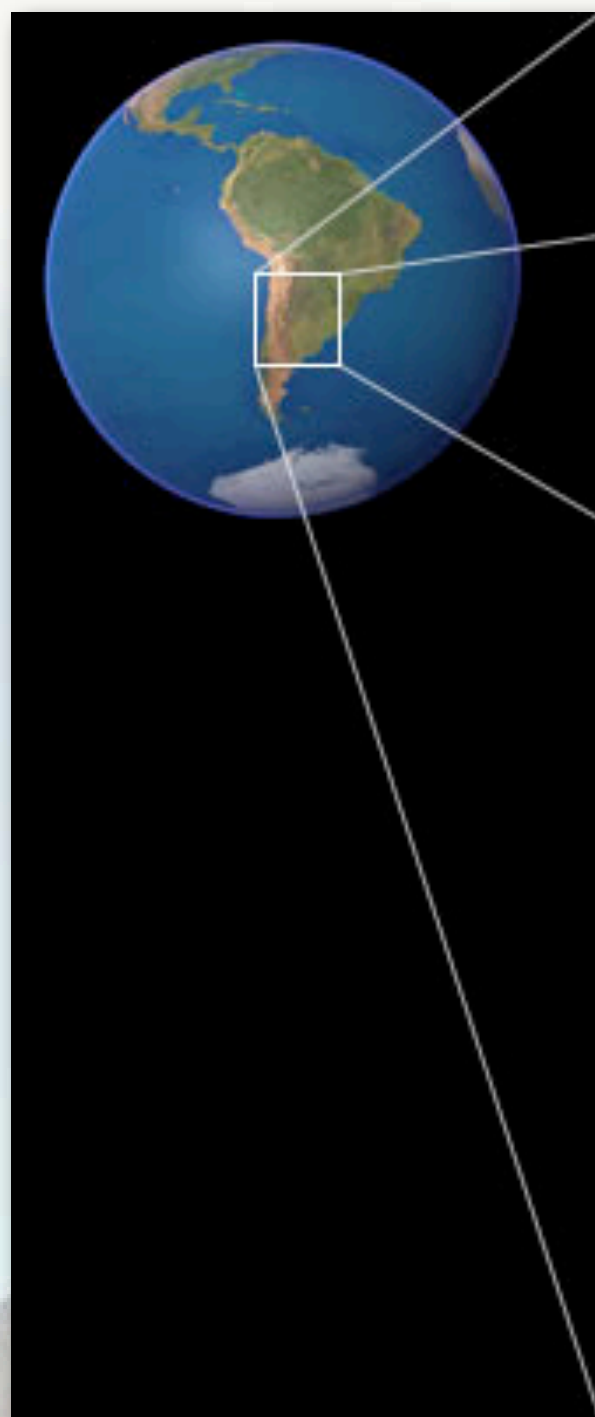


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- C_n^2 estimation with AFGL model (M1).
- C_n^2 estimation with Masciadri model (M2).
- Comparisons between C_n^2 models and two MM5 set up with SCIDAR.
- Estimation of Seeing and comparisons with DIMM and SCIDAR data in Paranal and Armazones.
- Conclusions and next steps.



Paranal Observatory : -24.4°S and -70.25°W , 2650 m.a.s.l.
Armazones site: -24.57°S and -70.18°W , 3000 m.a.s.l.



VLT at Paranal

ESO PR Photo 08/99 (5 December 1999)

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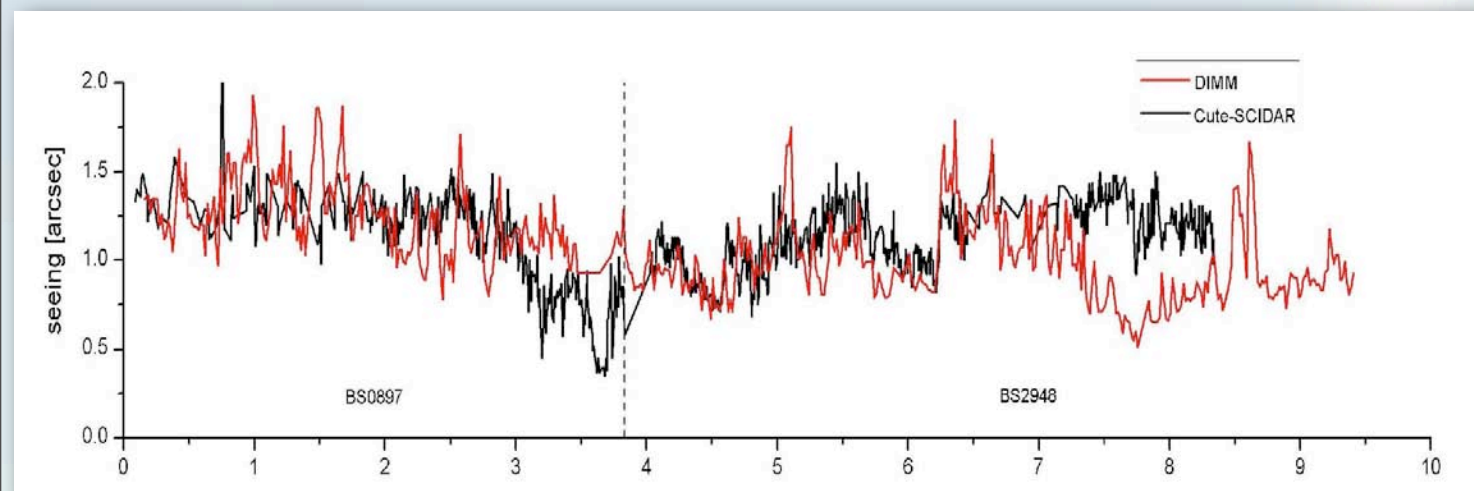
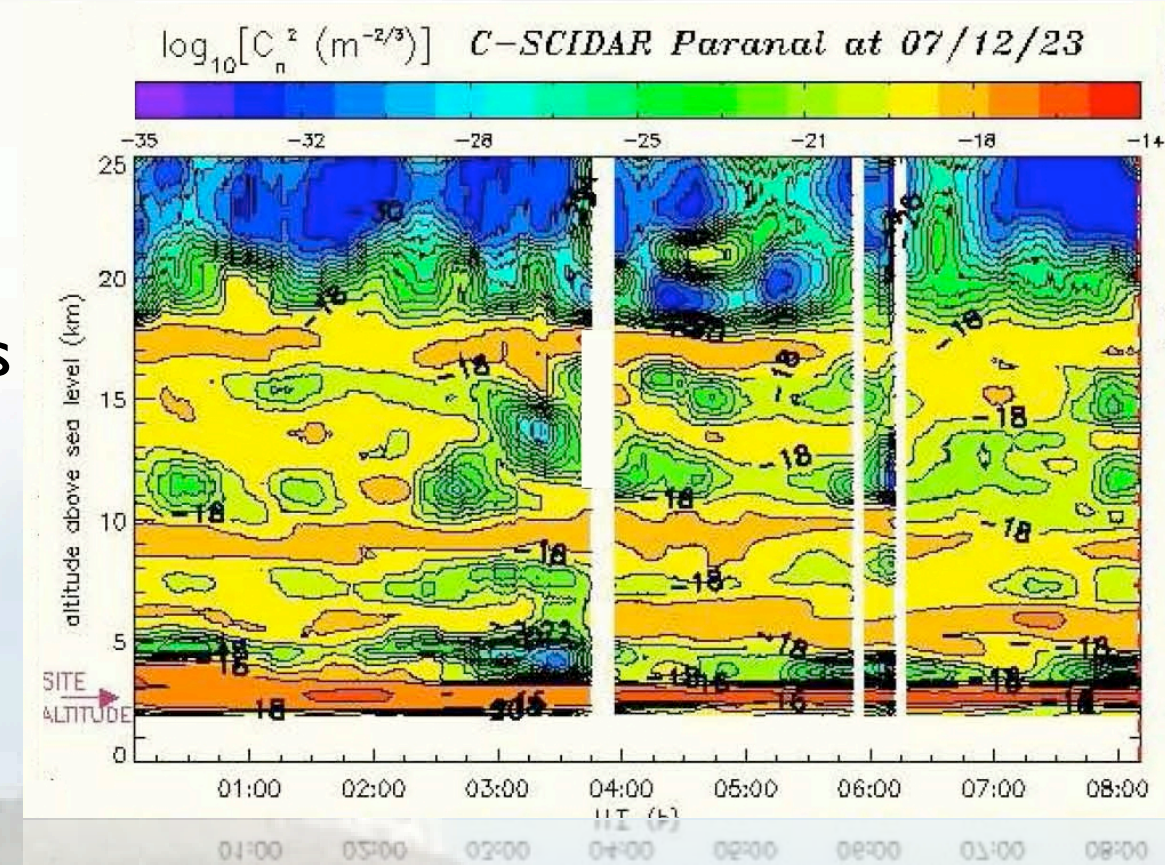
Local time: Friday 1 September 2001

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Campaign Measurement of OT (C_n^2) and seeing in Paranal 10 November to 24 of December of 2007

SCIDAR (Scintillation Detection and Ranging): It analyzes stellar scintillation of a binary star target produced by turbulent layers in the atmosphere (Vernin and Roddier, 1973). The vertical resolution is in layers of 300 m until 25 km.



DIMM (Differential Image Motion Monitor) measures fluctuations of the angular distance between two images produced by two circular apertures about 8 – 10 cm diameter separated by about 20 cm. Theory states that the rms image motion is proportional to the turbulence integral along the full light path in the atmosphere. So, the weighting function for DIMM does not depend on altitude. (Kornilov et al, 2006).



Methods to estimate the OT from meteorological data

Refraction index : C_n^2 Tatarski's equation

$$C_n^2 = 2.8 M^2 L^{4/3}$$

$$M^2 = \left[\left(\frac{-79 \times 10^{-6} P}{T^2} \right) \left(\frac{dT}{dz} + \gamma \right) \right]^2$$

T= Temperature (K) P= Pressure (mb) γ = dry adiabatic lapse rate

L is the “outer scale”, and a method to obtain it is proposal by AFGL (Dewan et al, 1993)

instability criteria:

$$Ri = N^2 / S^2 < 0.25$$

$$S \equiv \left[\left(\frac{dV_n}{dz} \right)^2 + \left(\frac{dV_e}{dz} \right)^2 \right]^{1/2}$$

$$L^{4/3}(h) = 10^{1.57+40S} \quad \text{troposphere}$$

$$L^{4/3}(h) = 10^{0.503+51.2S} \quad \text{stratosphere}$$

$$Y = 1.57 + 40S_{raw} \quad \text{troposphere}$$

$$Y = 0.503 + 51.2S_{raw} \quad \text{stratosphere}$$

$$C_n^2(z) = 2.8 (0.1)^{4/3} M^2 10^Y$$

AFGL Model: MI



M2: Masciadri et al, 2001

OT by Gladstone's law :

$$C_n^2 = \left(\frac{80 \times 10^{-6} P}{\theta^2} \right) C_\theta^2$$

Temperature fluctuations in
a L path

$$C_\theta^2 = 0.59 L^{4/3} \left(\frac{d\theta}{dz} \right)^2 \phi_3$$

ϕ : thermal and dynamic
stability in the
atmosphere = 0.78

$$L = \sqrt{\frac{2E}{g \frac{d\theta}{dz}}}$$

L in function of the TKE (E)
and Potential Temperature (θ)

θ : Potential temperature

using L in C_θ^2 equation's we obtain:

$$C_\theta^2 = 0.20 \cdot \theta^{2/3} \left(\frac{d\theta}{dz} \right)^{4/3} E^{2/3} \phi_3$$

Finally using C_θ^2 we obtain OT

$$C_n^2 = 3.35 \times 10^{-6} P^{2(1-2R/c_p)} \theta^{-10/3} \left(\frac{d\theta}{dz} \right)^{4/3} E^{2/3}$$



To obtain the input data for C_n^2 models have been used the MM5 meteorological mesoscale model configured to Paranal observatory and Armazones E-ELT

4 domain centered in Paranal at -24.4° and -70.25°

Domain 1 : 80x80 grid points every 27 km

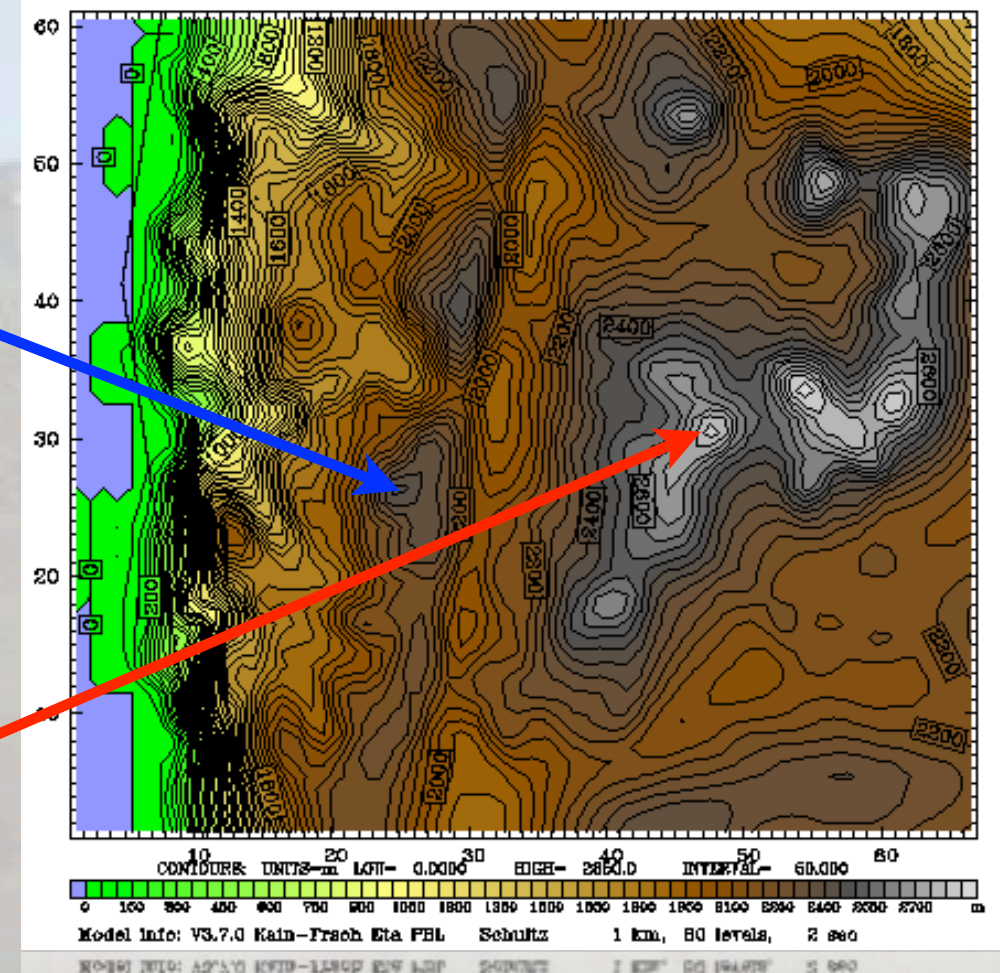
Domain 2 : 70x79 grid points every 9 km

Domain 3 : 82x82 grid points every 3 km

Domain 4 : 61x67 grid points every 1 km

Paranal site
elevation: 2650
m.a.s.l.

Armazones site
elevation: 3000
m.a.s.l.





MM5 set up

1. Boundary conditions from GFS to 18 UTC:

1.25°x1.25° horizontal resolution (~100 km)

12 pressure levels, 1000 mb to 100 mb

2. Integration of 80 sigma levels.

3. Parameterization:

Explicit moisture schemes:	Schultz
Cumulus schemes:	Kain-Fritsch
Planetary boundary layer:	ETA Gayno-Seaman
Atmospheric radiation:	RRTM

4. Time step:

Domain 1	60s	output 10 min
Domain 2	20s	output 10 min
Domain 3	6.67s	output 10 min
Domain 4	2.22s	output 10 min



Two configurations of MM5 model for TKE

Explicit moisture schemes:	Schultz
Cumulus schemes:	Kain-Fritsch
Planetary boundary layer:	ETA
Atmospheric radiation:	RRTM

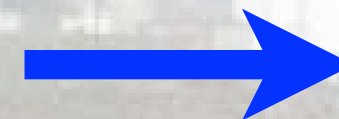


M1

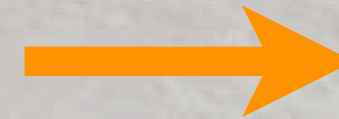


M2

Explicit moisture schemes:	Schultz
Cumulus schemes:	Kain-Fritsch
Planetary boundary layer:	Gayno-Seaman
Atmospheric radiation:	RRTM



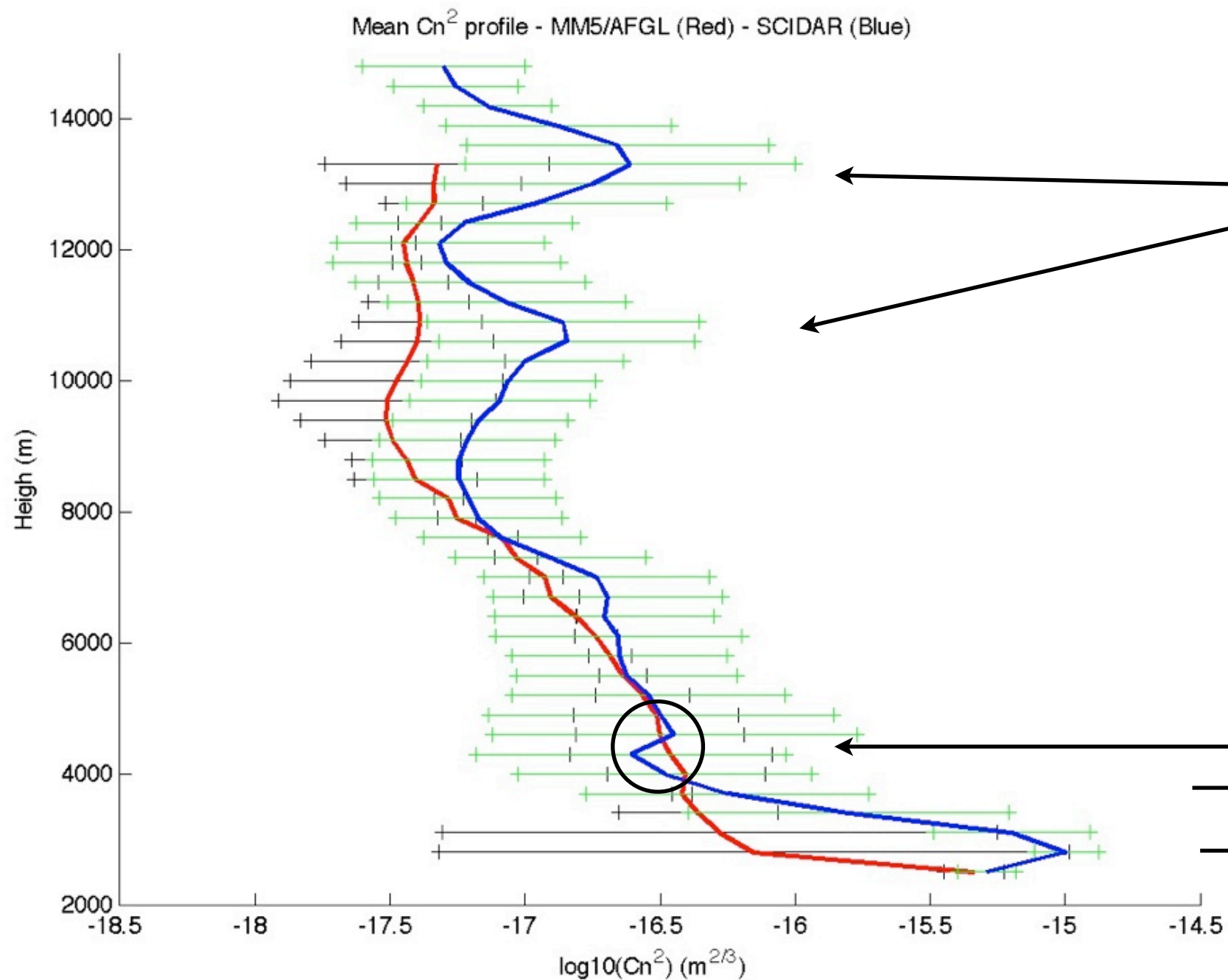
M1



M2



C_n^2 MI model MM5/ETA: 14 nights mean between Nov- Dec 2007



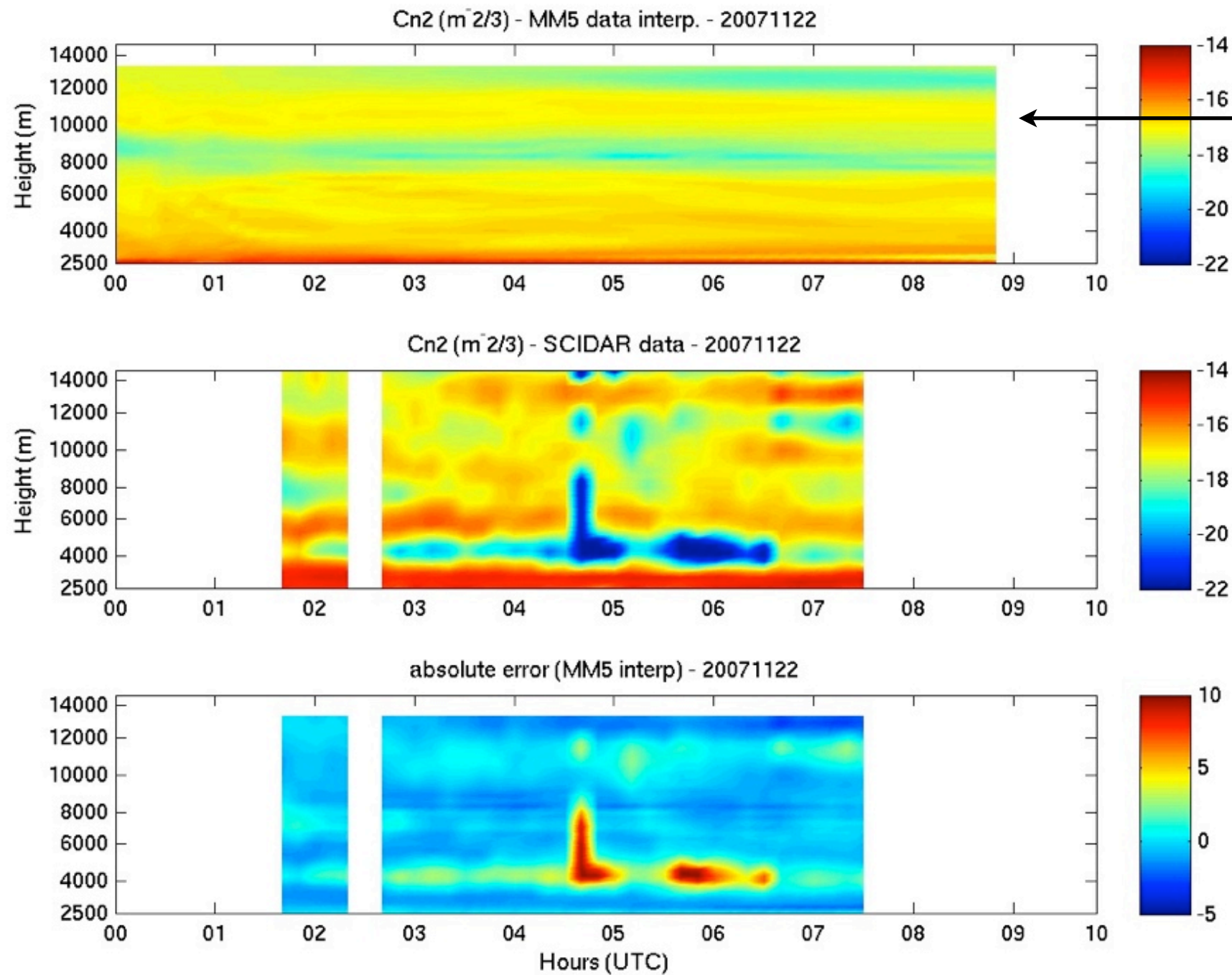
Jet stream effects

stability layer, it's true?

a lot of OT in the
boundary layer



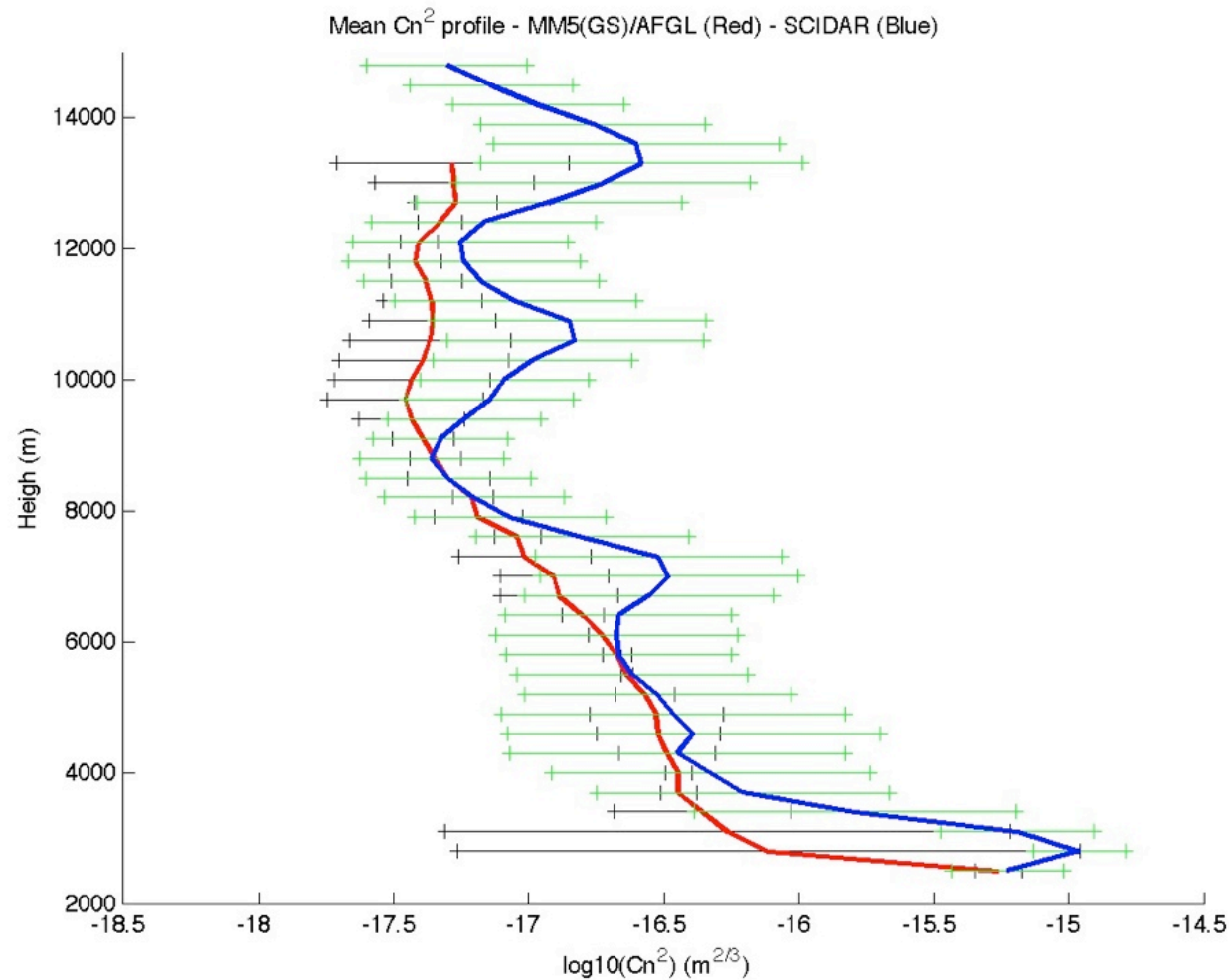
22-Nov-2007 night



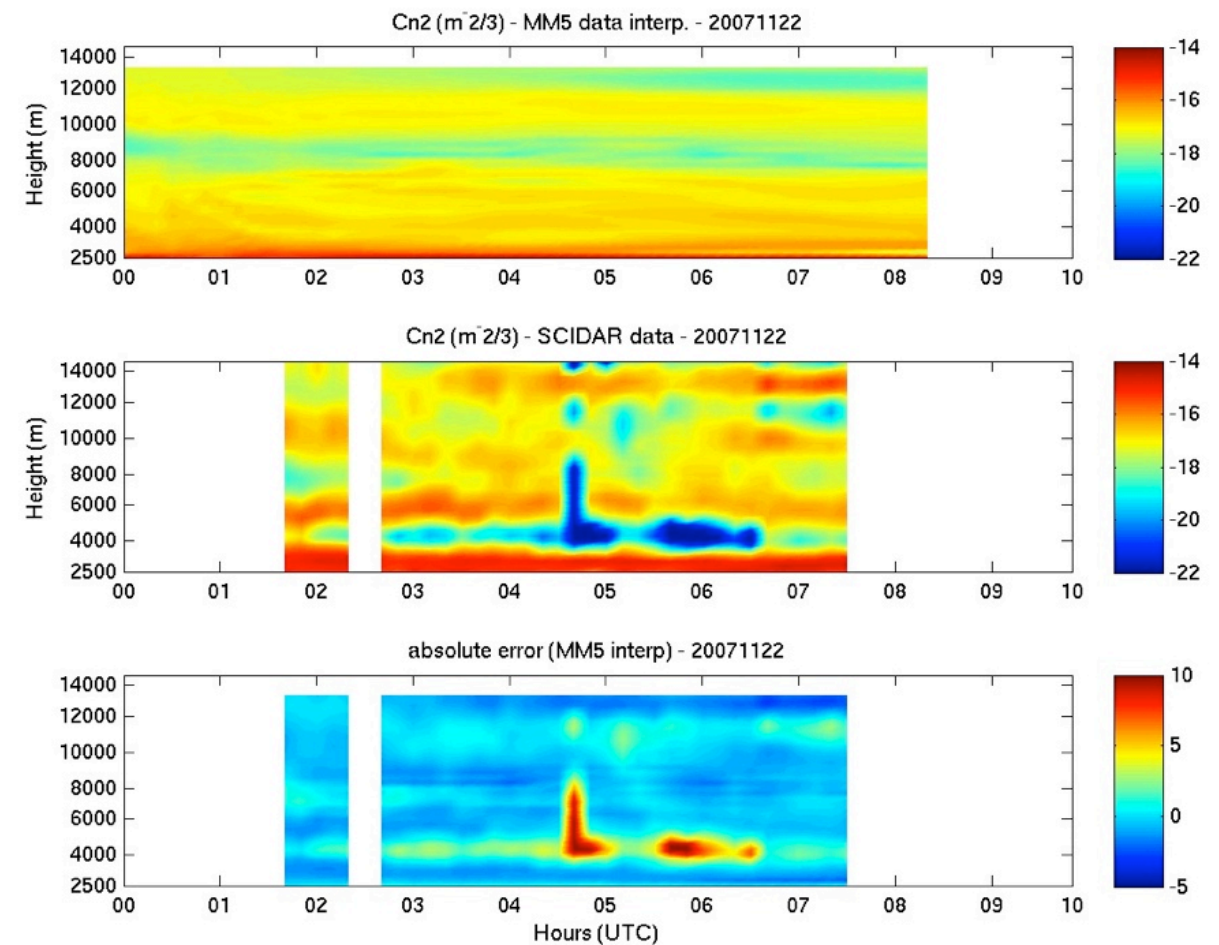
Jet stream effects



C_n^2 MI model MM5/GS: 14 nights mean between Nov- Dec 2007

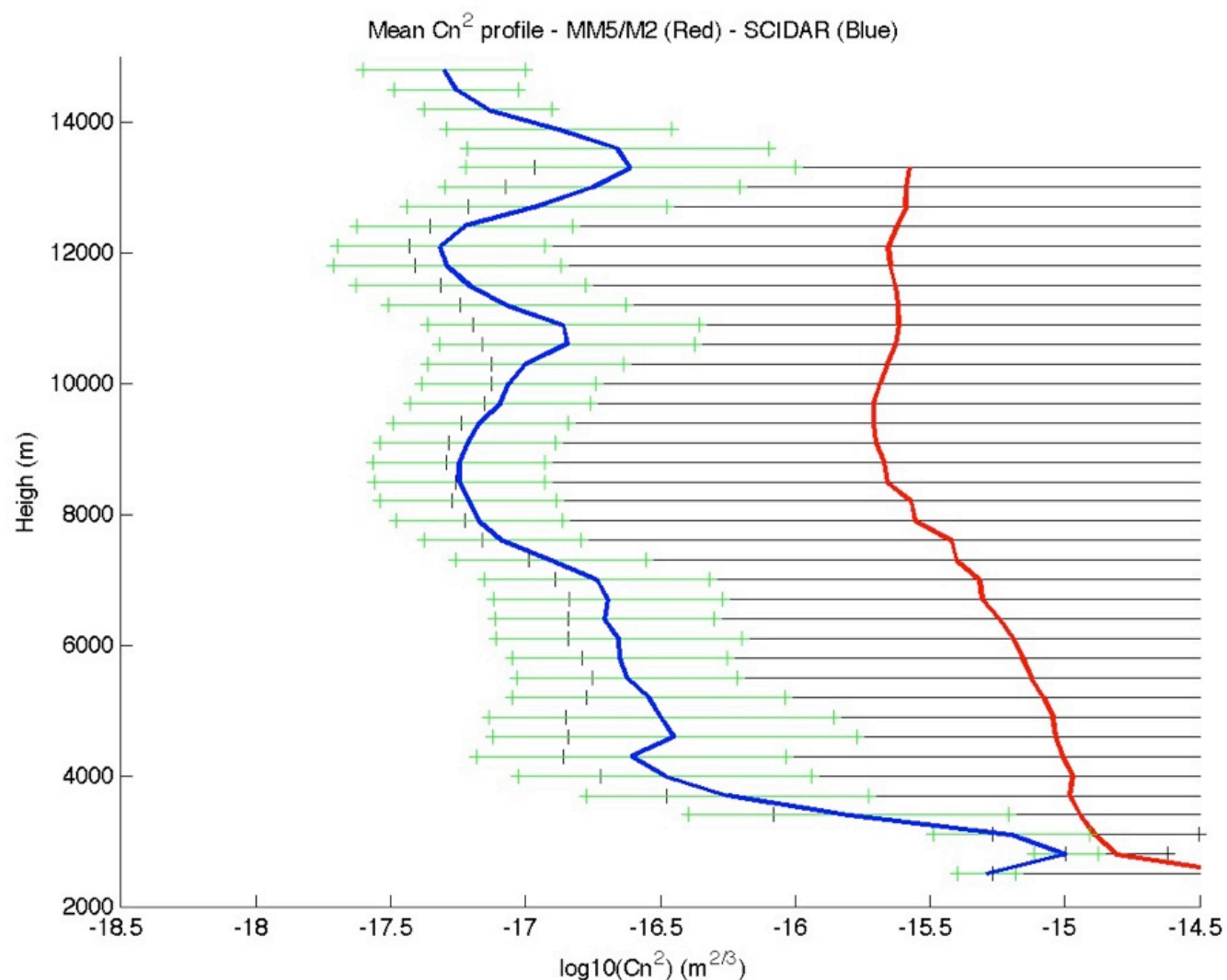


22-Nov-2007 night MM5/GS



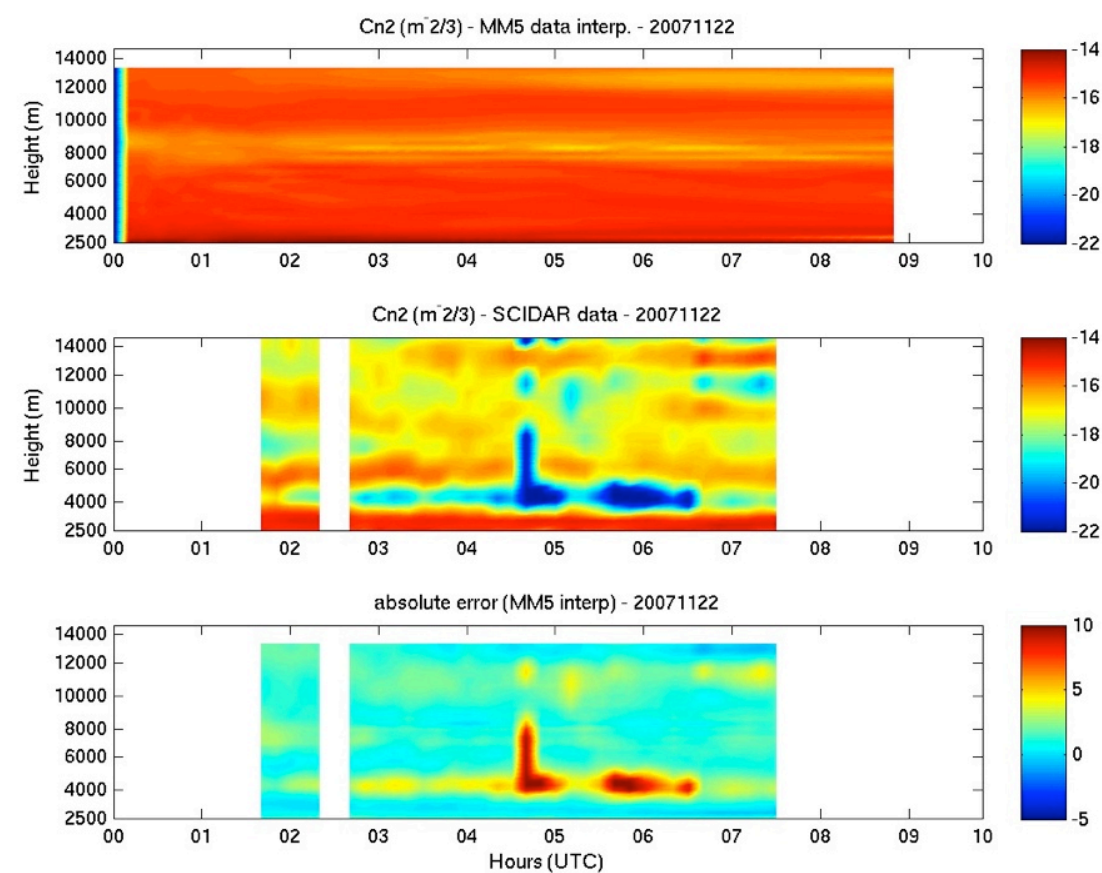


M2/ETA



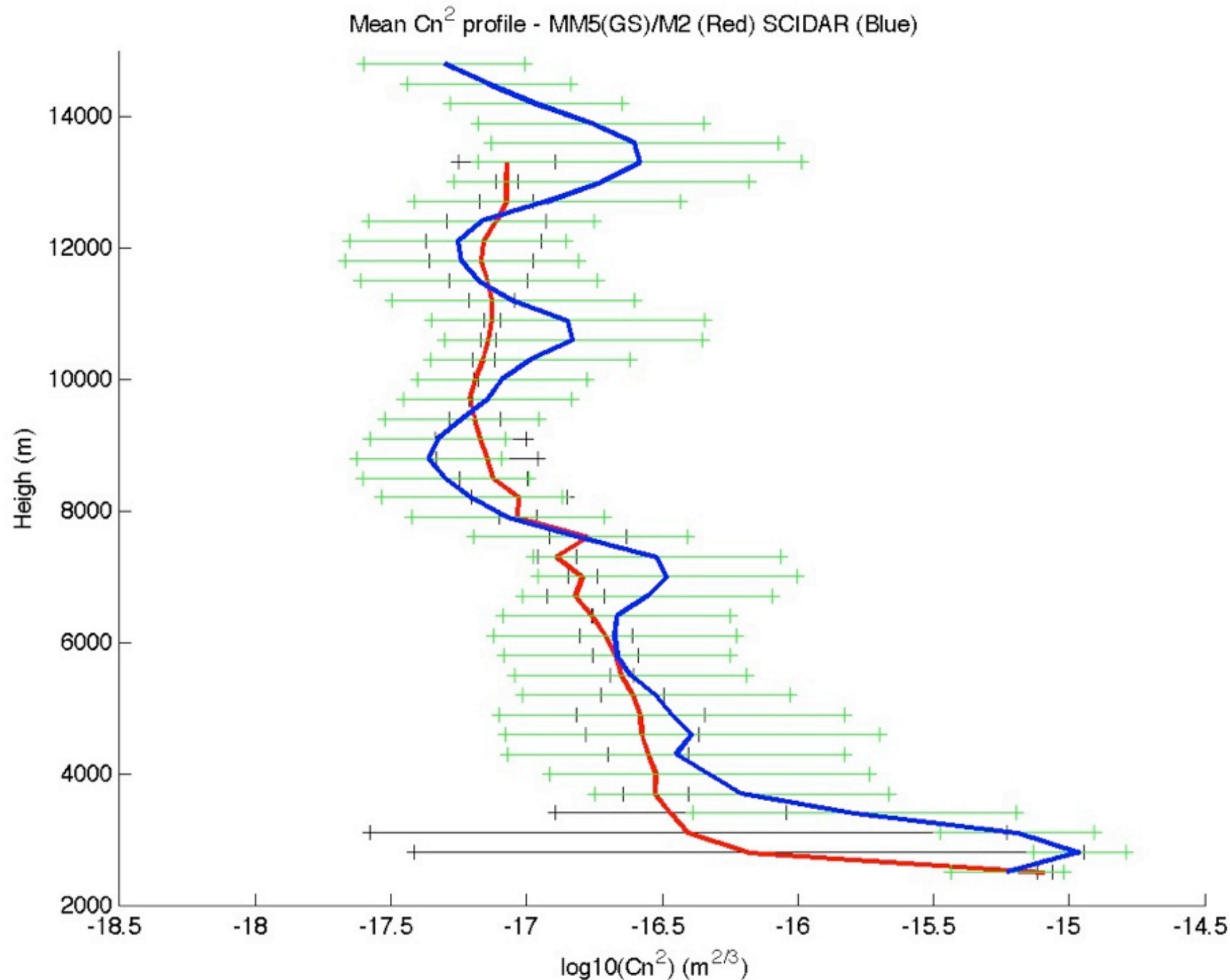
C_n^2 M2 model: 14 nights mean
between Nov- Dec 2007
MM5/ETA PBL

22-Nov-2007 night MM5/GS





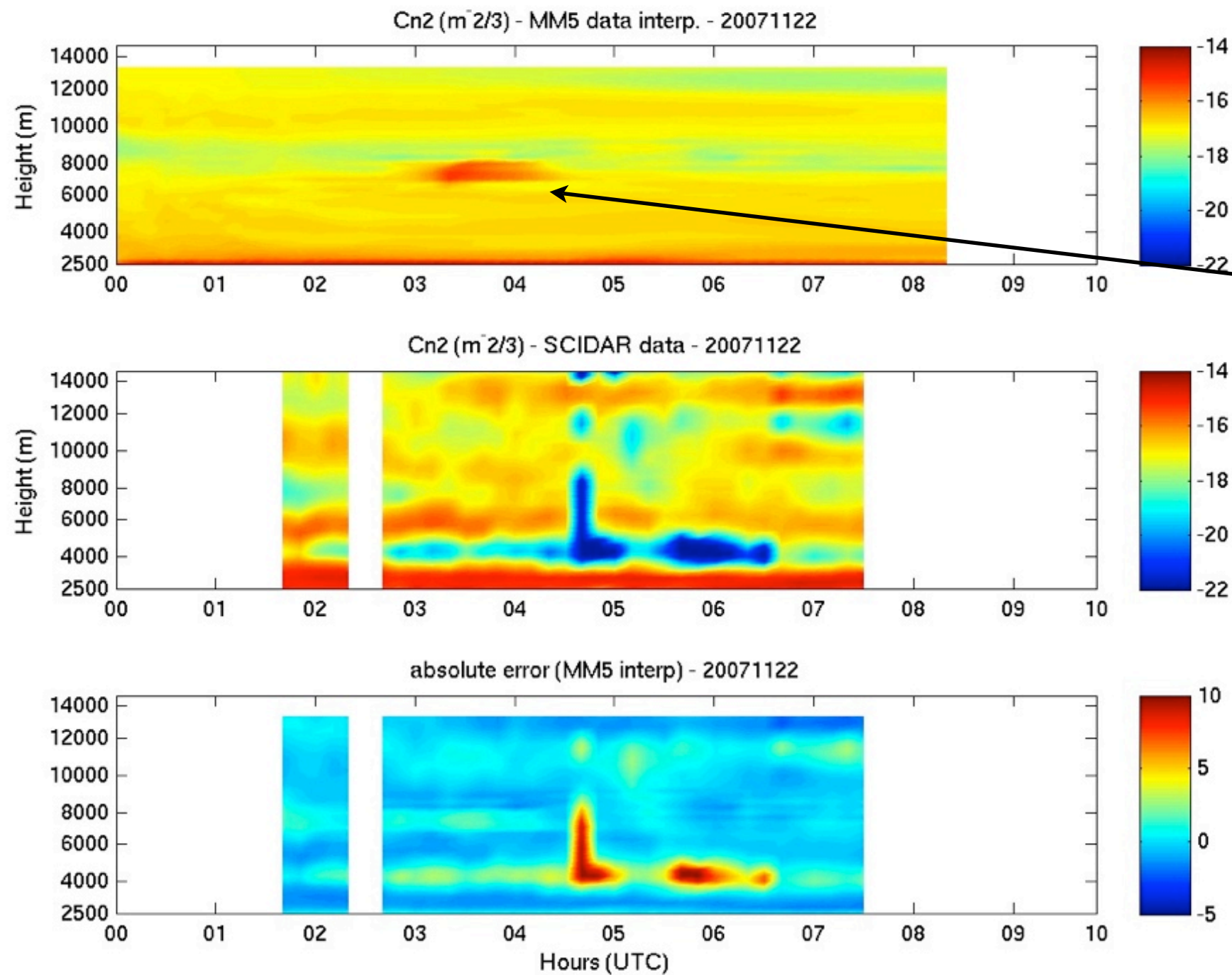
C_n^2 M2 model MM5/GS: 14 nights mean between Nov- Dec 2007



Error is slow in
comparison with the
MI model



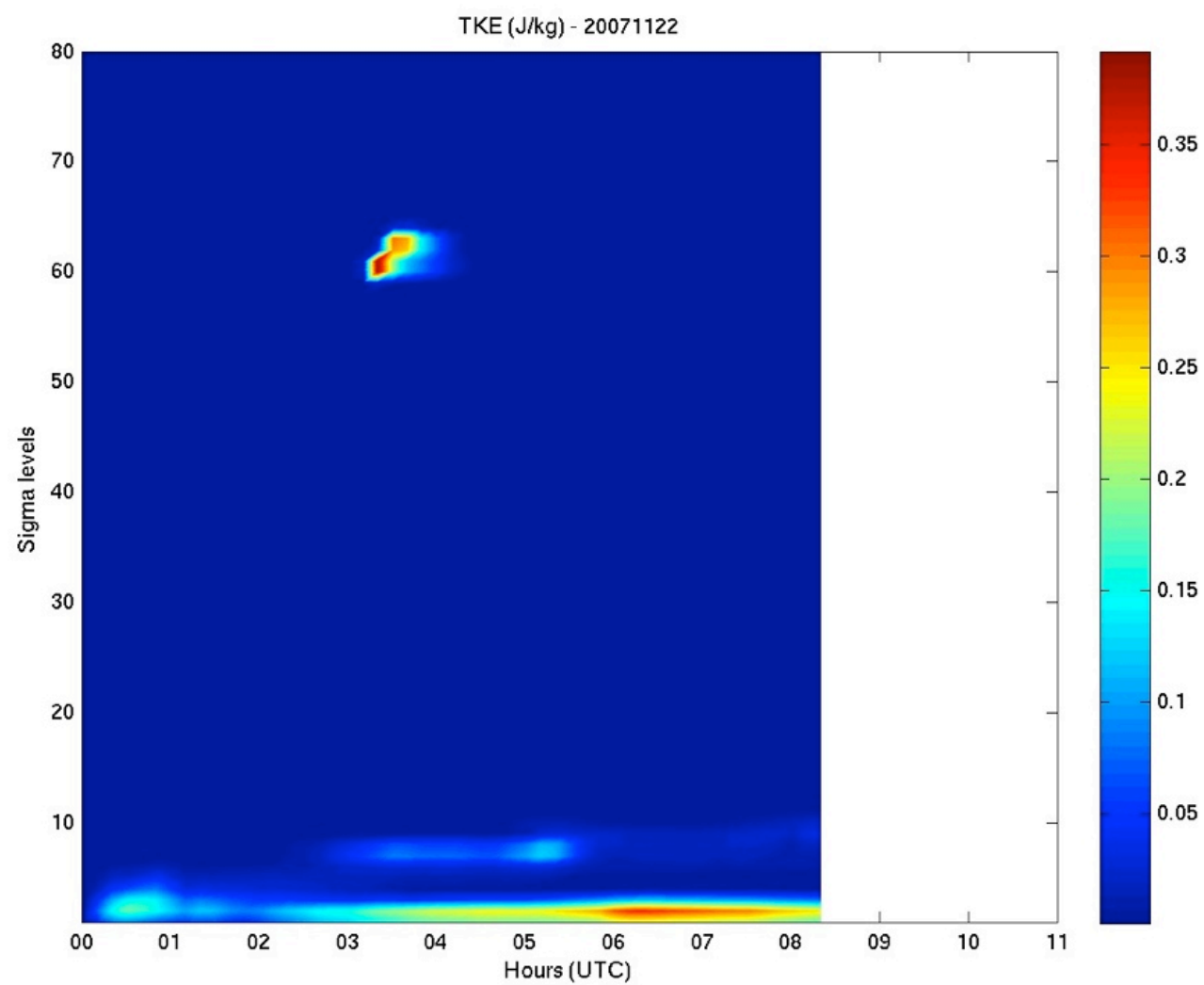
22-Nov-2007 night MM5/GS on M2 C_n^2



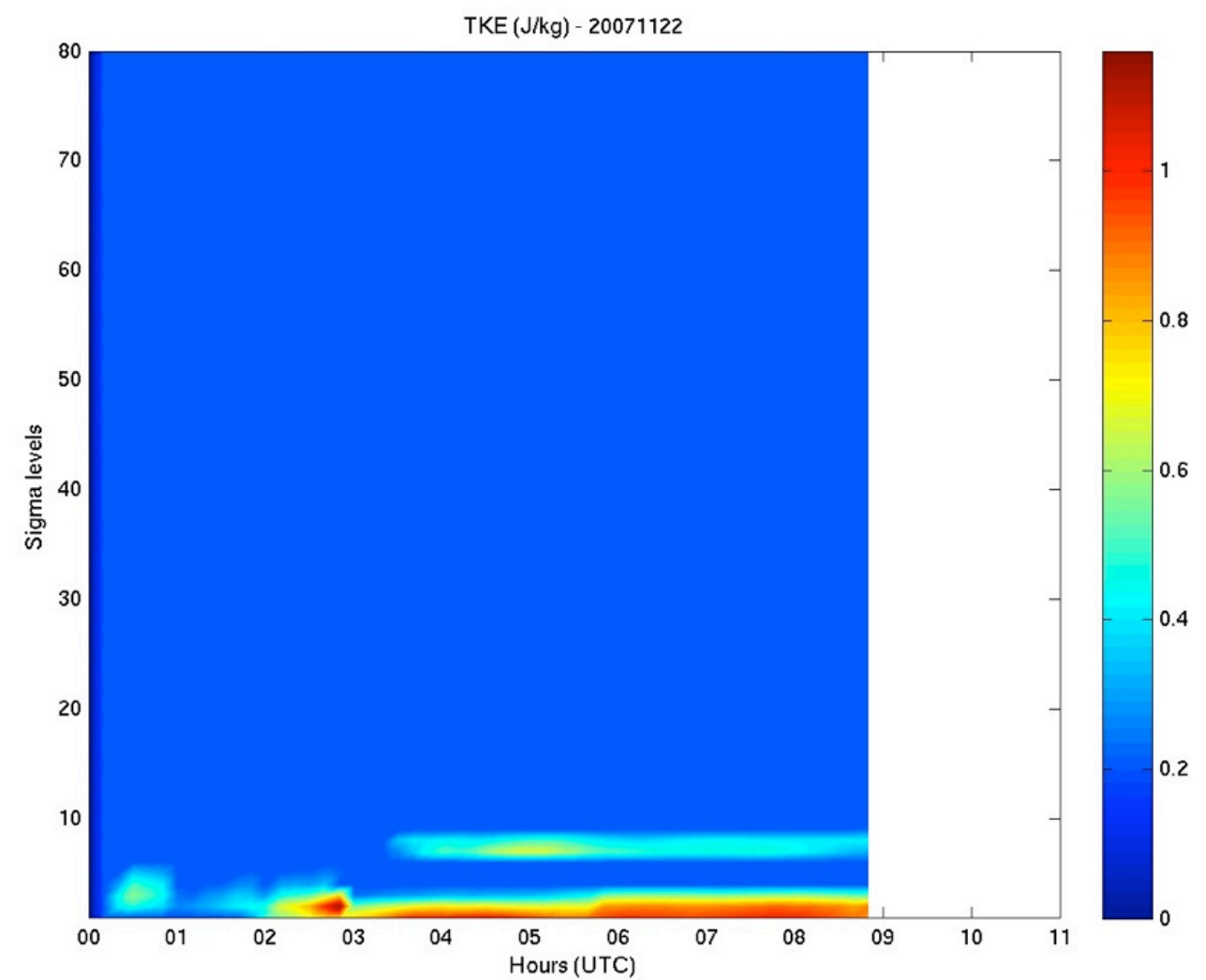
OT layer



TKE MM5/GS PBL 22-Nov-2007



TKE MM5/ETA PBL 22-Nov-2007





Seeing calculation

The seeing (ϕ) is other index of optical turbulence and represent the quality of the image in the telescope. The seeing is defined as the full width at half maximum of a star image at the focus of large diameter telescope. For to calculate the seeing first we need will obtain the Fried number or coherence length r_0 which is an important parameter in optical astronomy. The r_0 is obtained by vertical integration of the refractive index structure coefficient C_n^2 along the optical path L:

$$r_0 = \left[0.423 \left(\frac{2\pi}{\lambda} \right)^2 \int_0^L C_n^2(z) dz \right]^{\frac{-3}{5}} \text{ (m)}$$

where λ is the wavelength at which the telescope is observing (0.5×10^{-6} m).

Finally the seeing equation is:

$$\phi = 0.98 \frac{\lambda}{r_0}$$

The astronomers using the measurement in arcsec (“)



Classification of Seeing in Paranal

Range (arcsec)	Type	Adaptive Optic (OT)
$< 0.5''$	Super	System off adaptive optics, maximum diffraction, excellent image quality with normal instruments
$0.5'' < 0.8''$	Good	Adaptive Optic works well, infrared instruments (20 microns) as VISIR accomplishes diffraction limits.
$0.8'' < 1.2''$	Default	
$1.2'' <$	Bad	The adaptive optics begins to have problems correcting the atmosphere

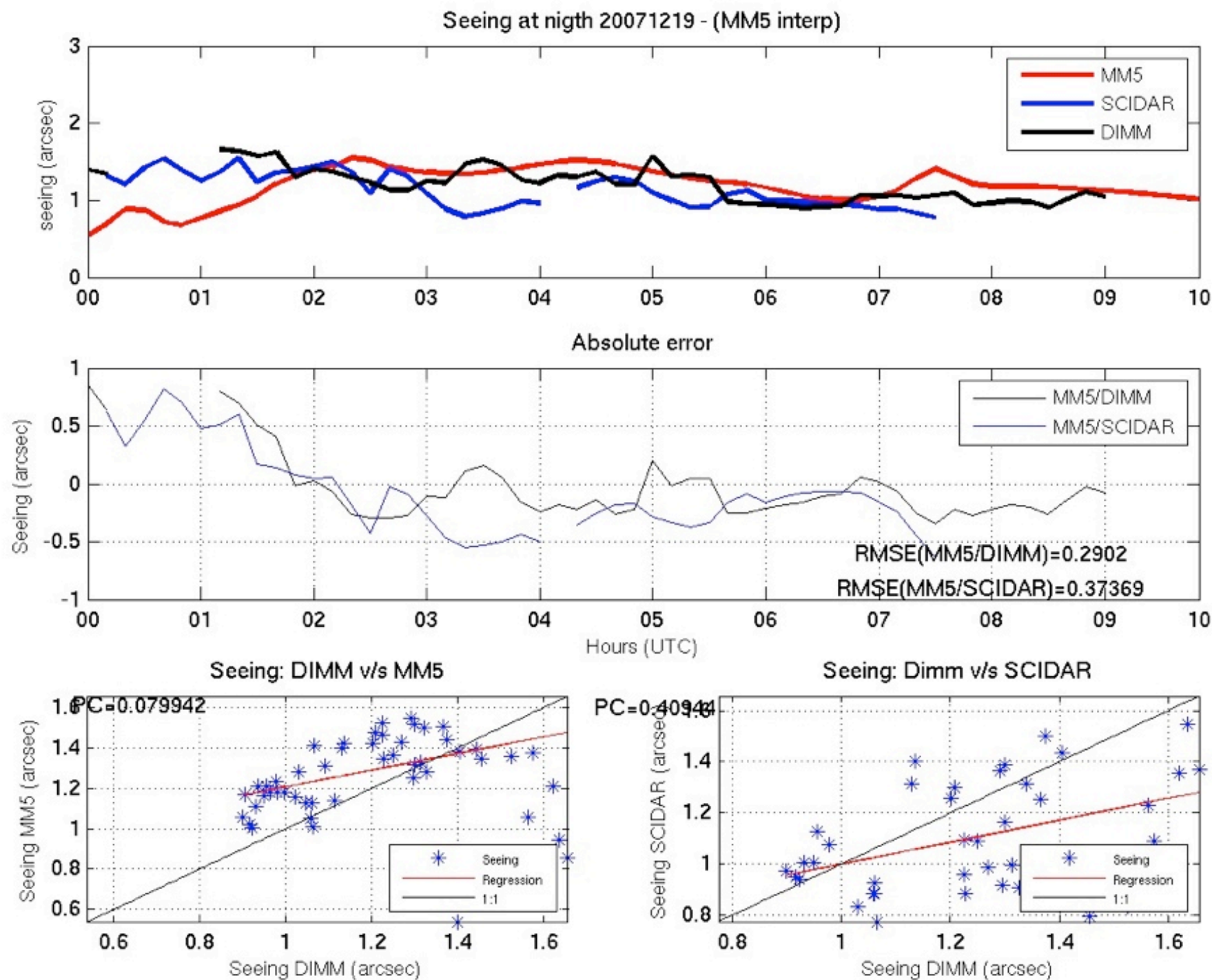
By Julio Navarrete from Paranal Observatory

www.eso.org



Seeing with MI and MM5/ETA PBL

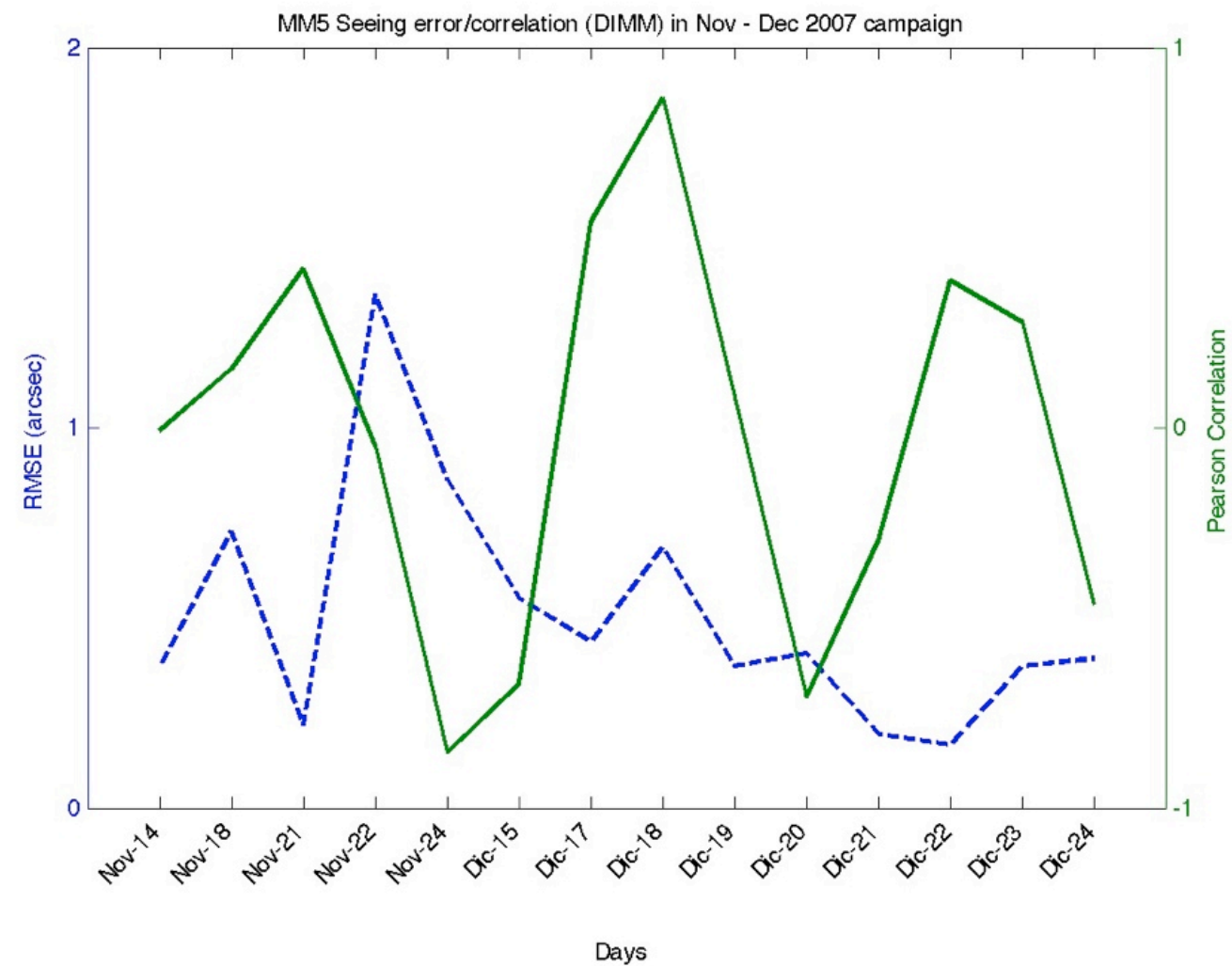
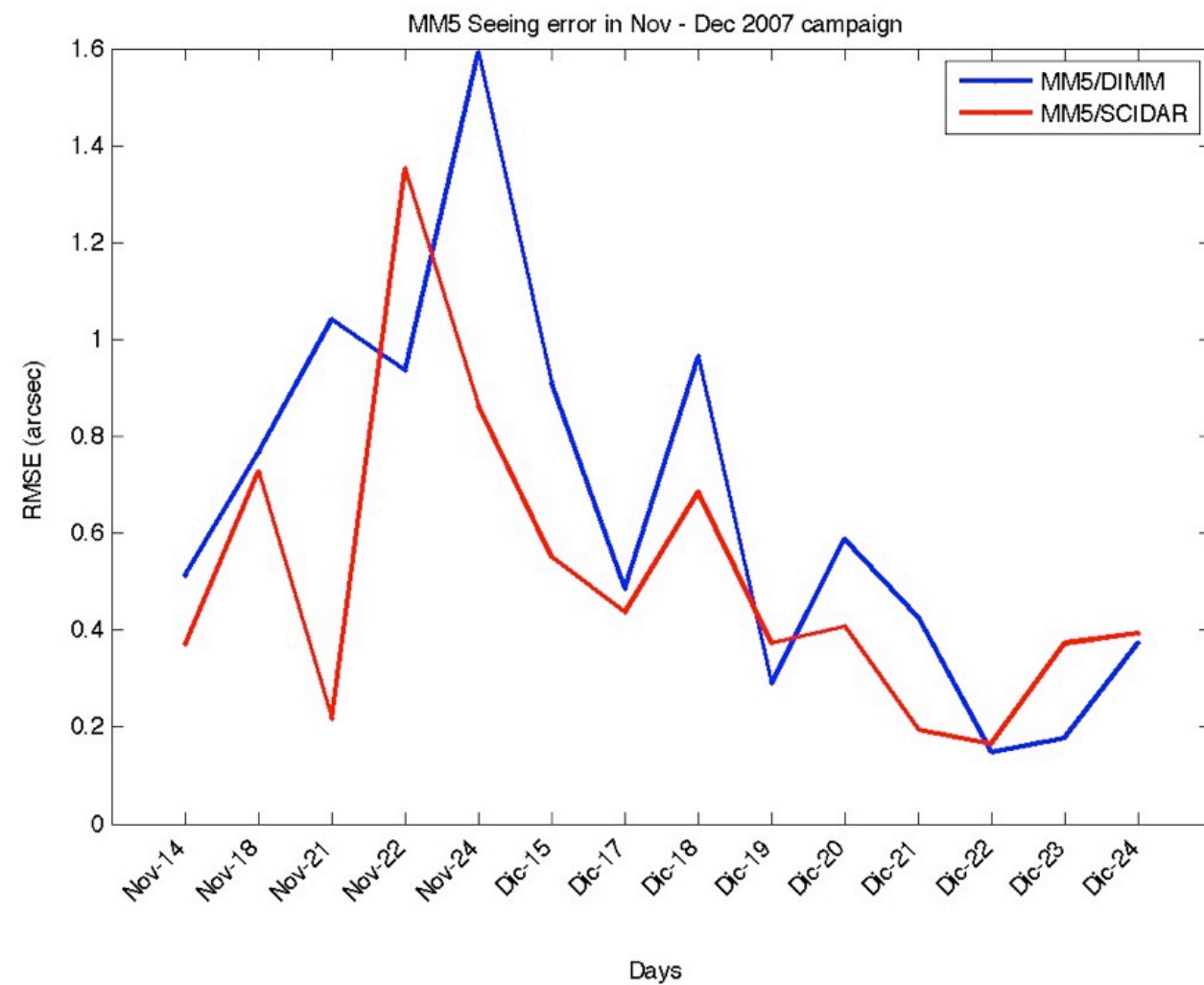
19-12-2007
Low error





MI - MM5/ETA

Pearson's correlation and RMSE with DIMM

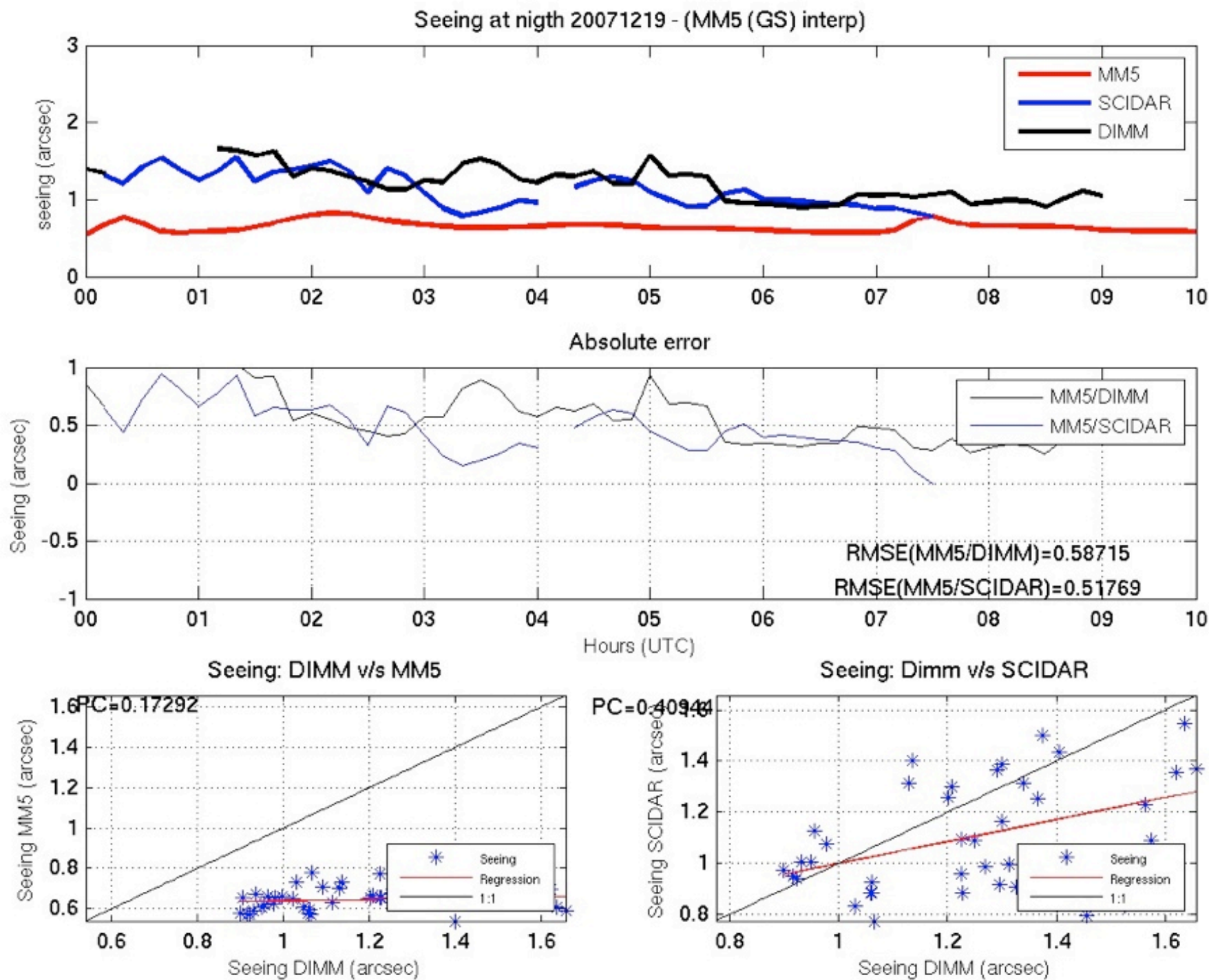


RMSE with DIMM and SCIDAR



Seeing with MI and MM5/GS PBL

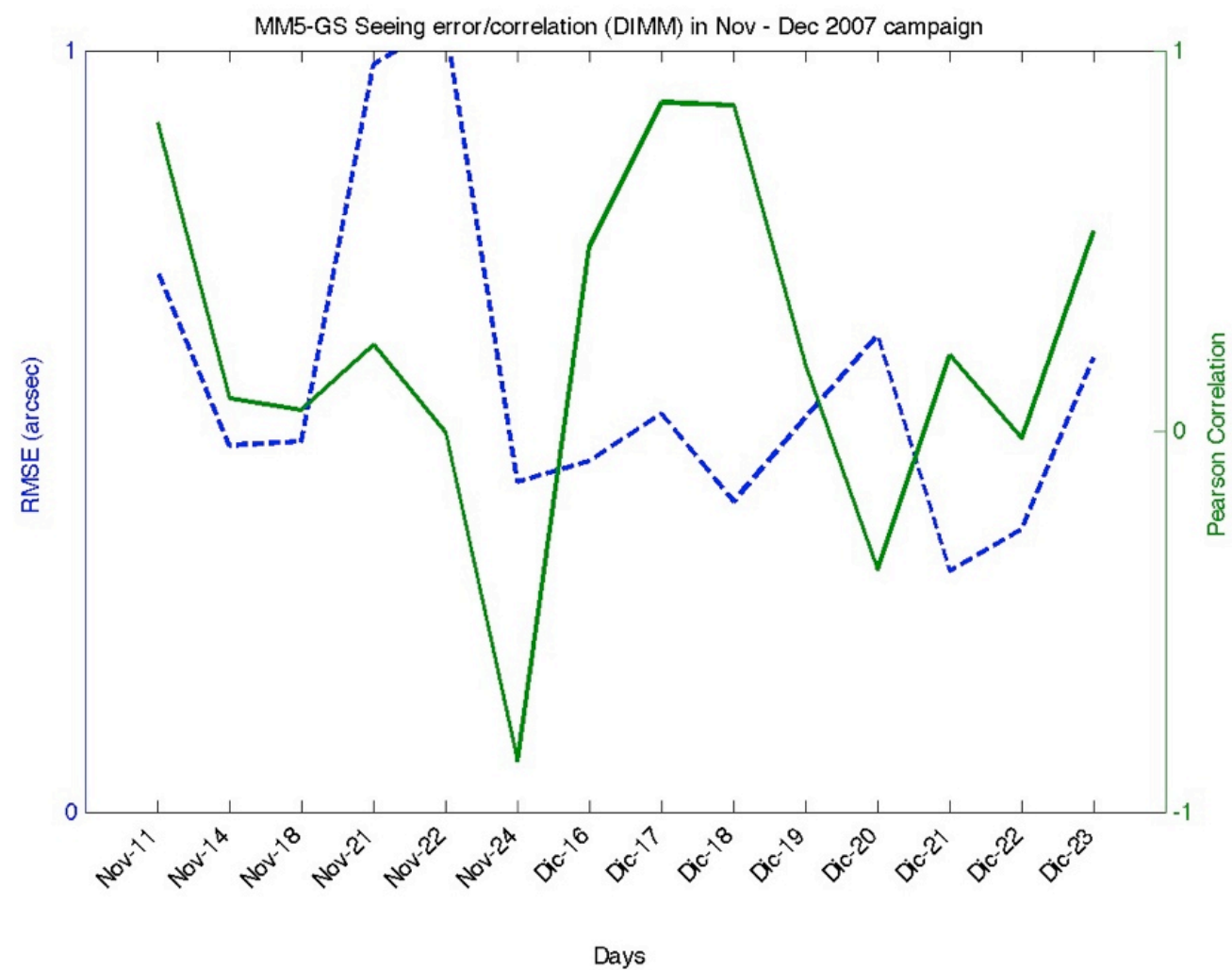
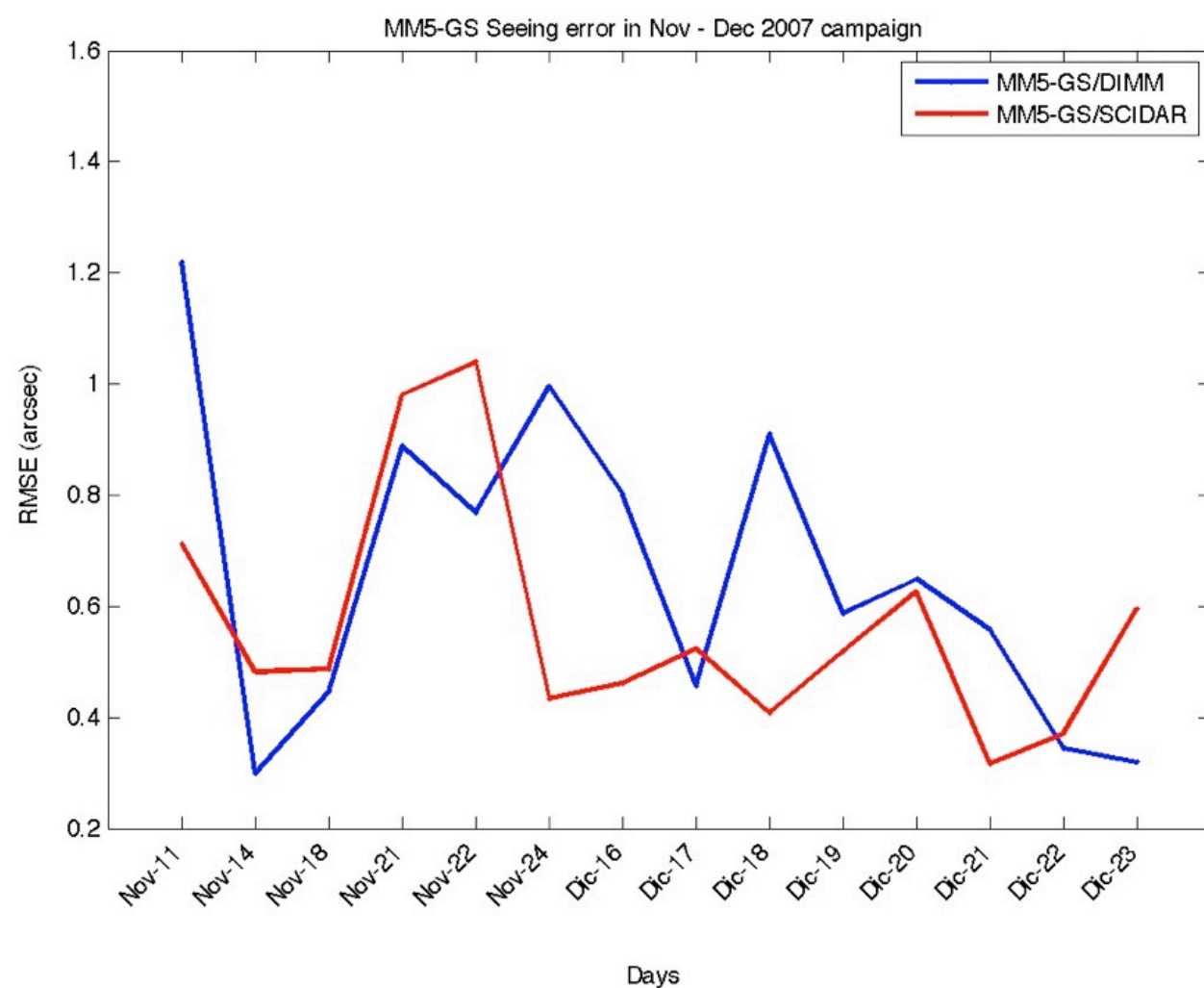
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MI - MM5/GS

Pearson's correlation and RMSE with DIMM

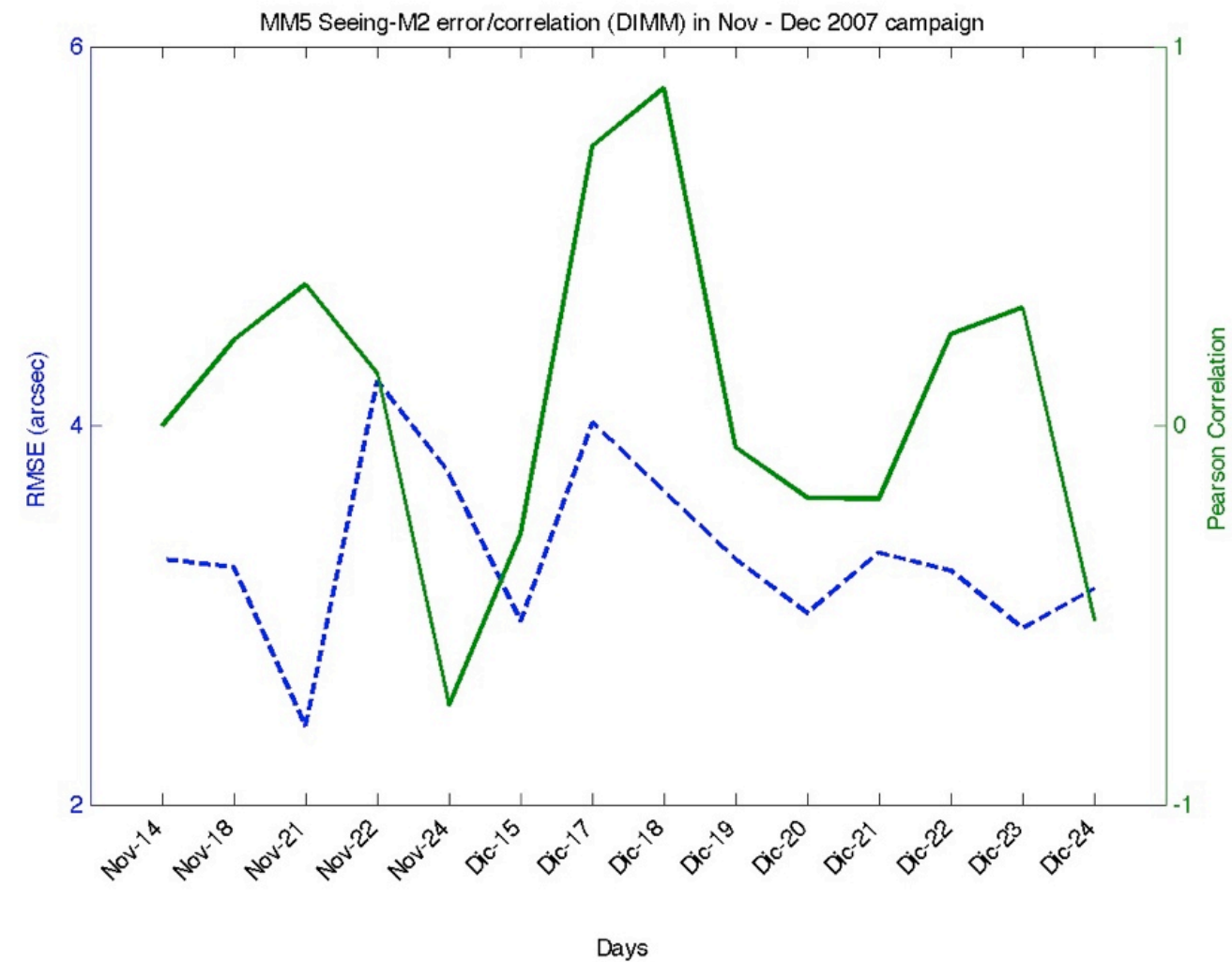
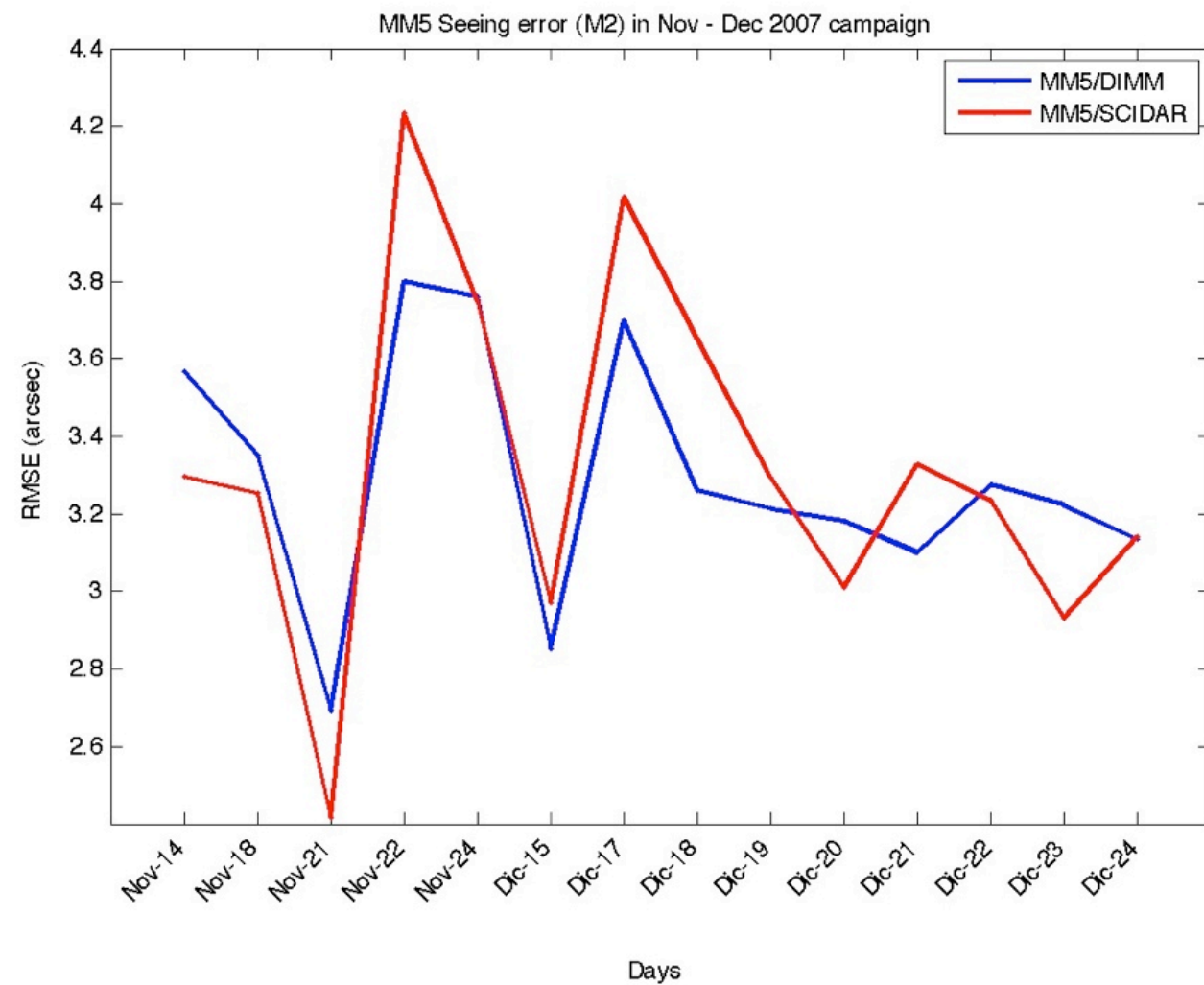


RMSE with DIMM and SCIDAR



Seeing with M2 and MM5/ETA PBL

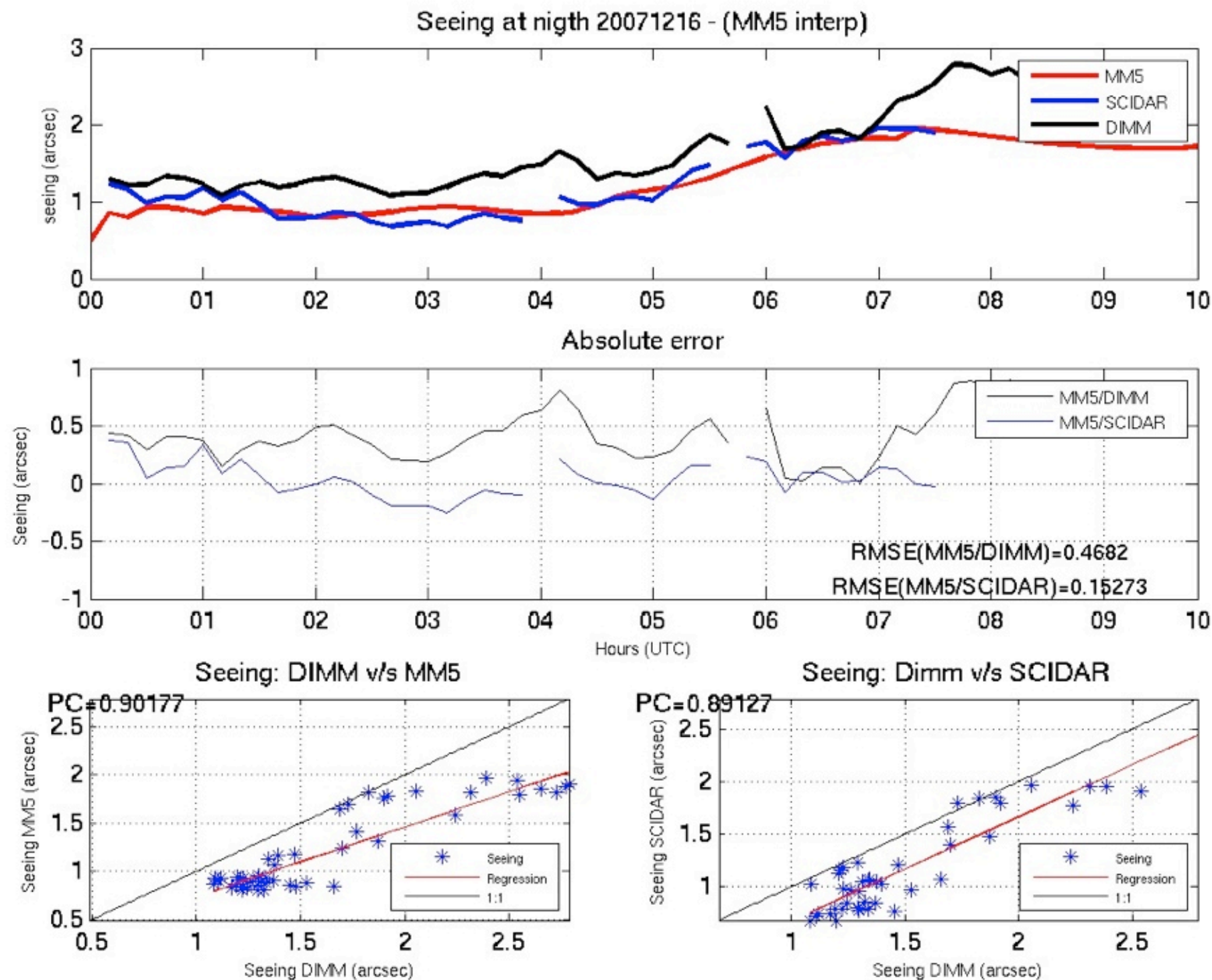
Pearson's correlation and RMSE
with DIMM



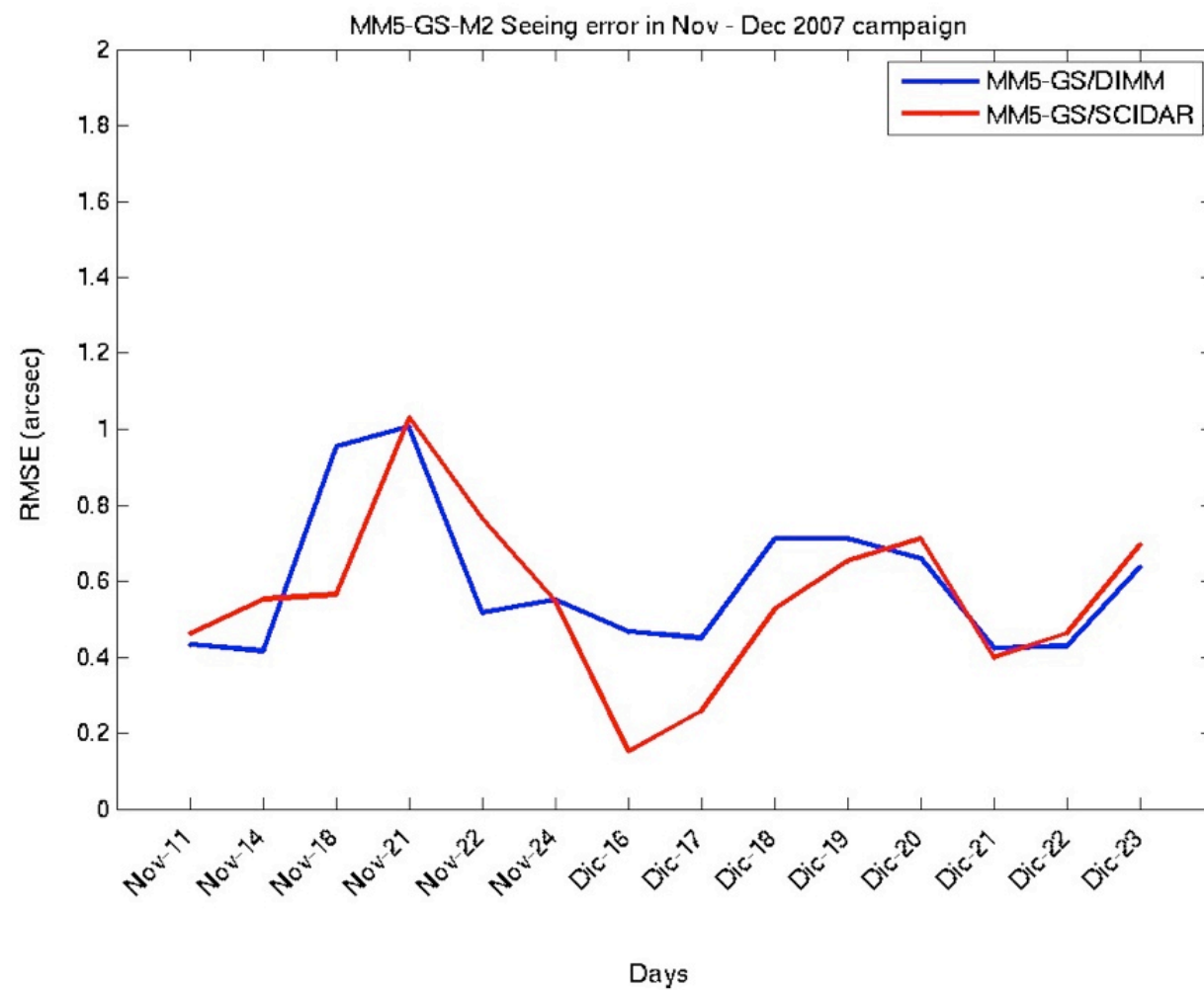
RMSE with DIMM and SCIDAR



Seeing with M2 and MM5/GS PBL

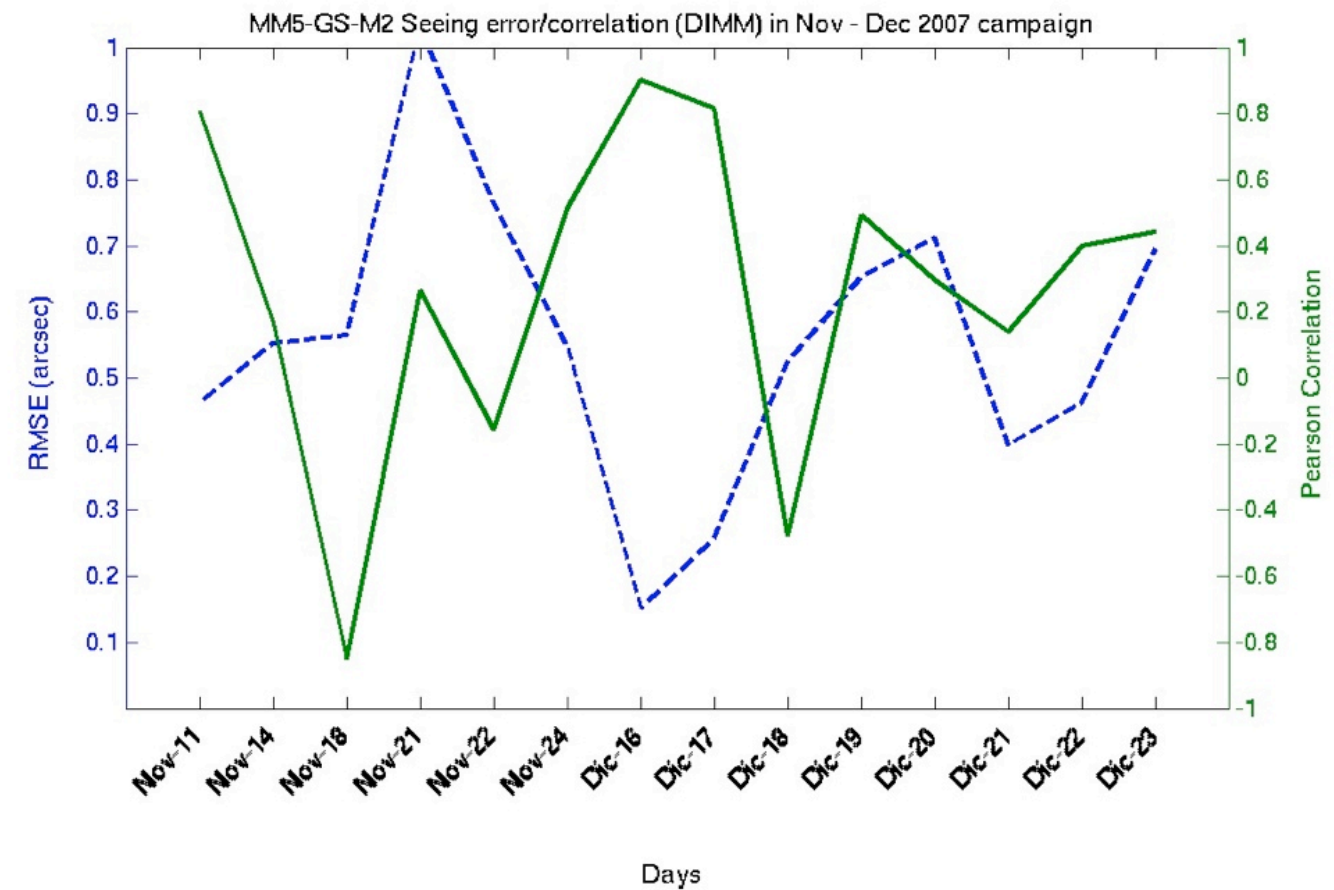


16-12-2007
Low error



Seeing with M2 and MM5/GS PBL

Pearson's correlation and RMSE
with DIMM



Root mean square error with DIMM
and SCIDAR

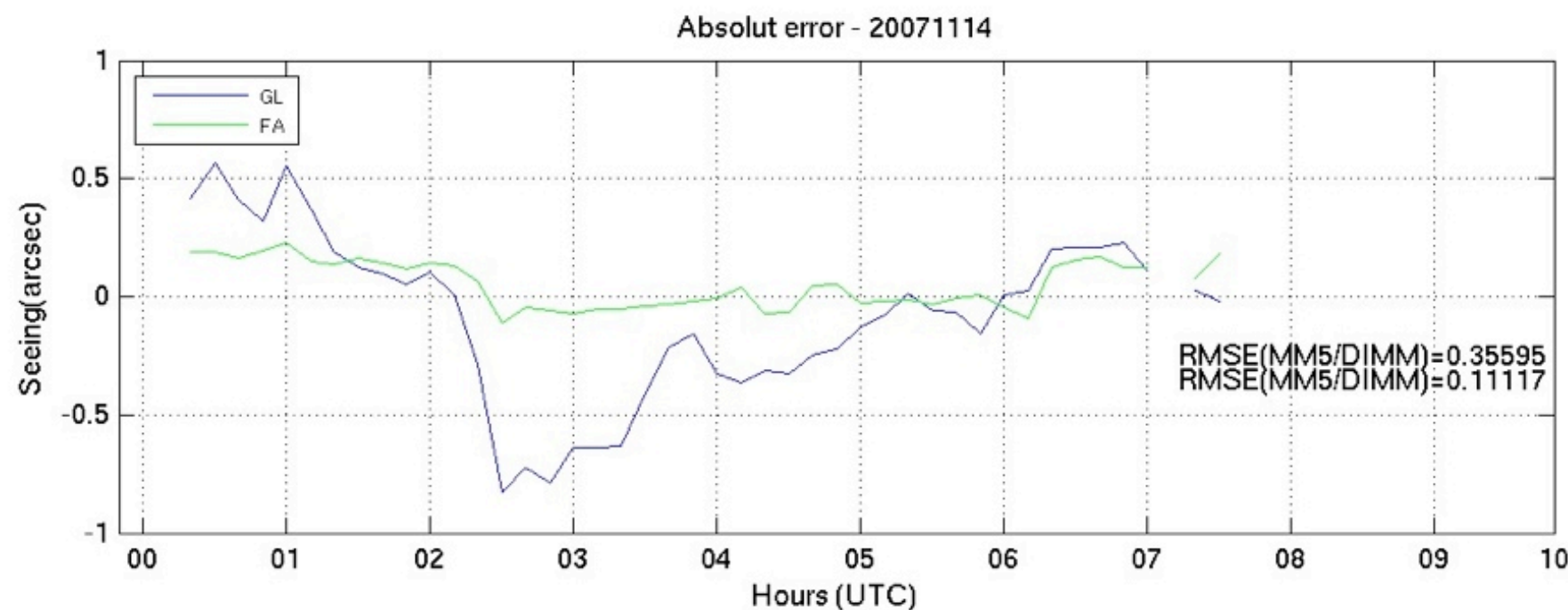
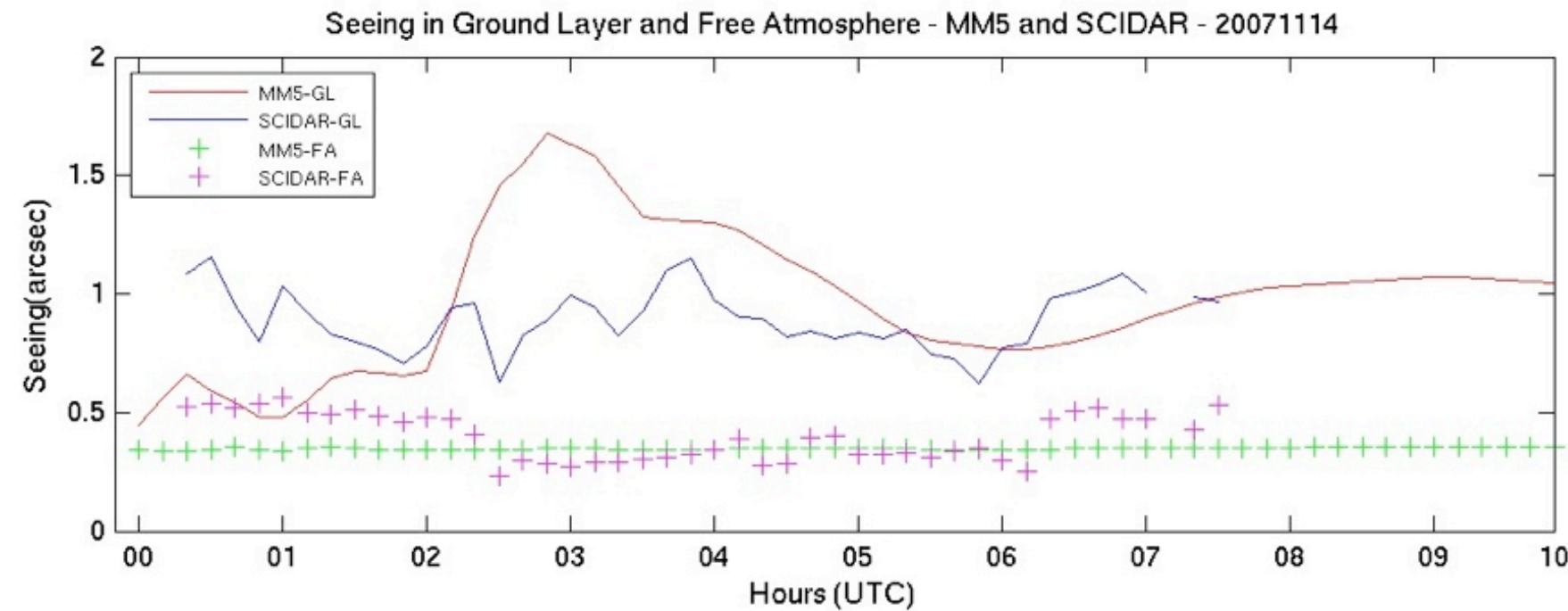


Seeing: Ground Layer (GL) - Free Atmosphere (FA) AFGL model - MM5/ETA

14-12-2007

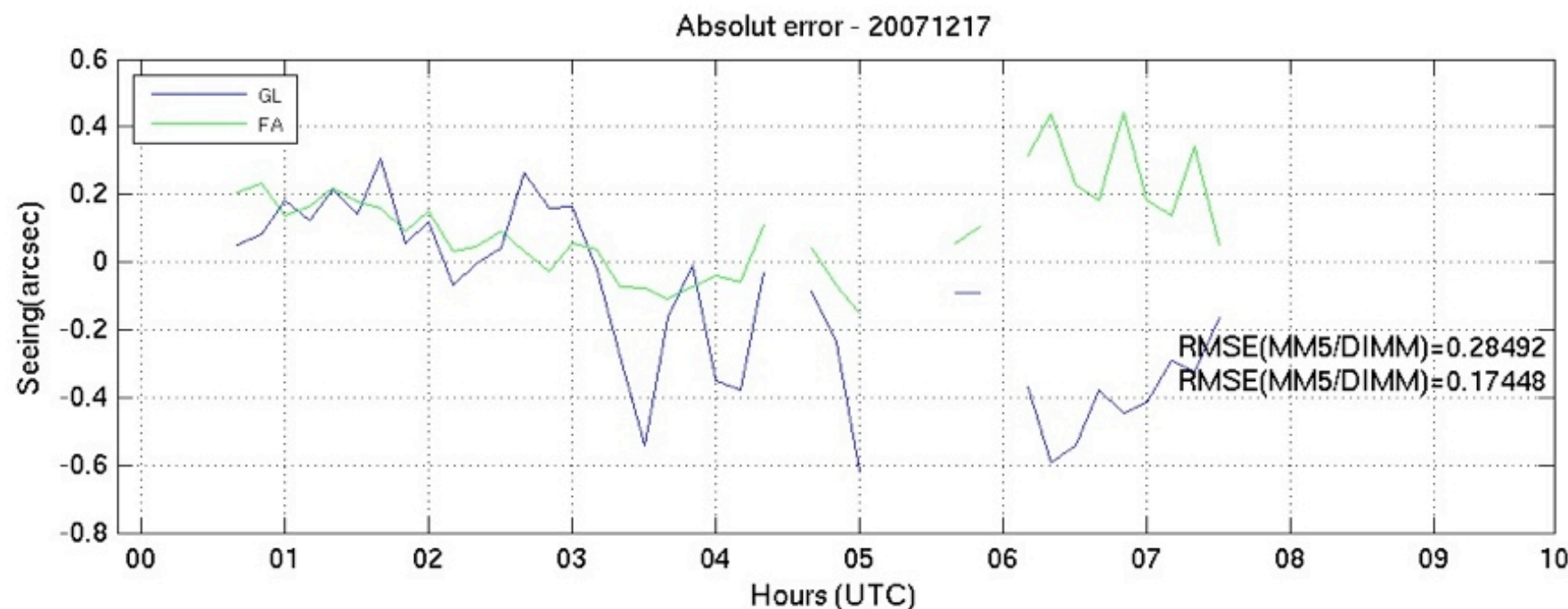
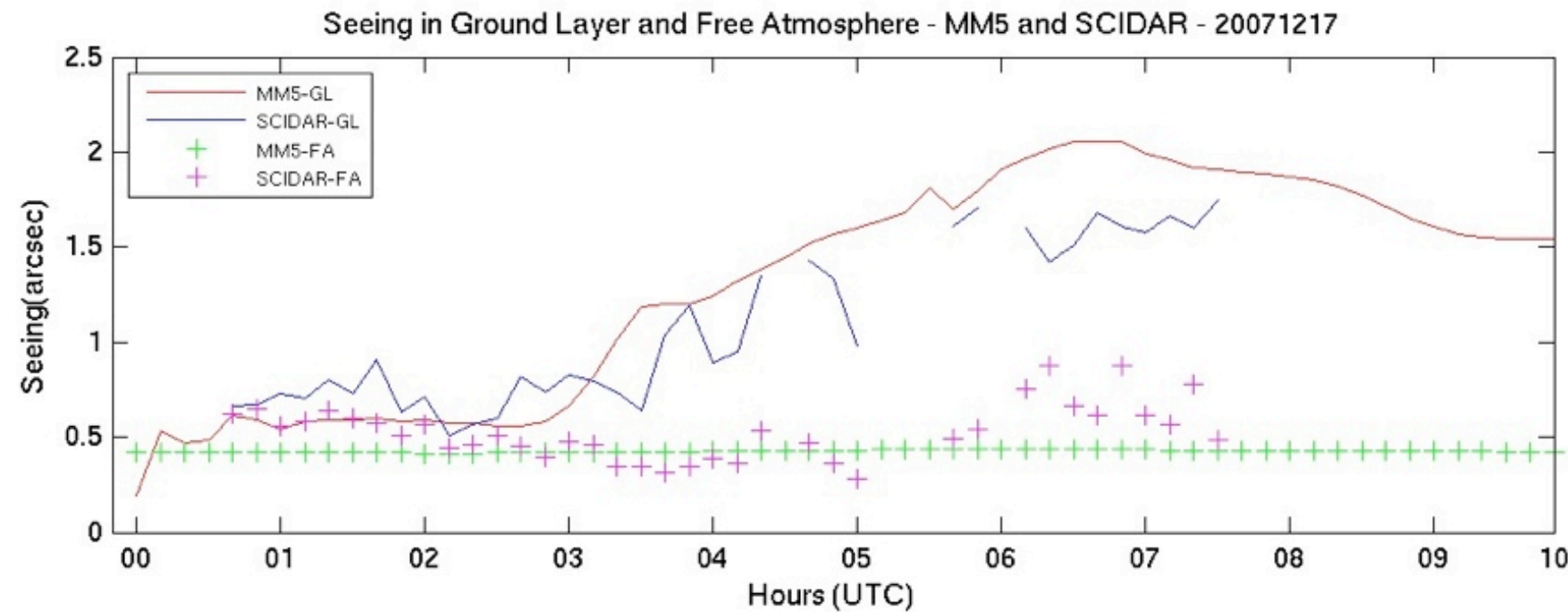
GL < 1 km

FA > 1 km





Seeing: Ground Layer (GL) - Free Atmosphere (FA) M2 model - MM5/GS

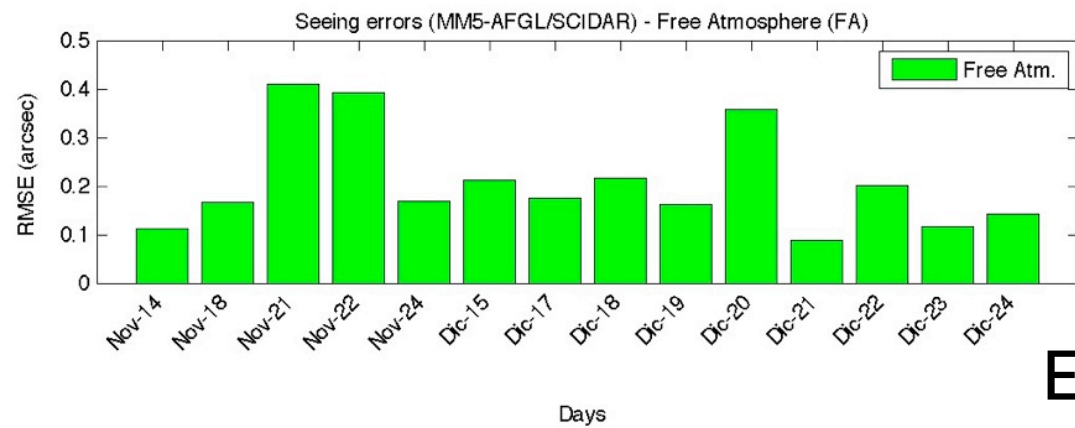
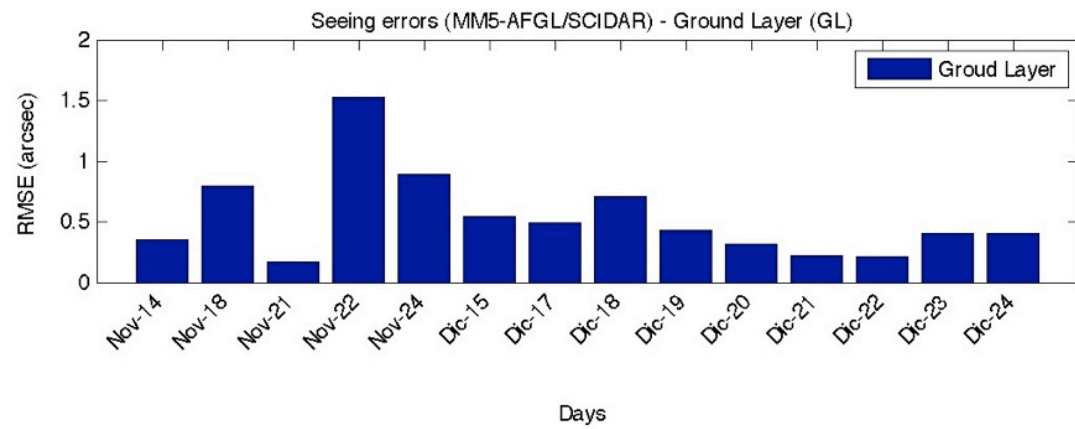


17-12-2007

GL < 1 km

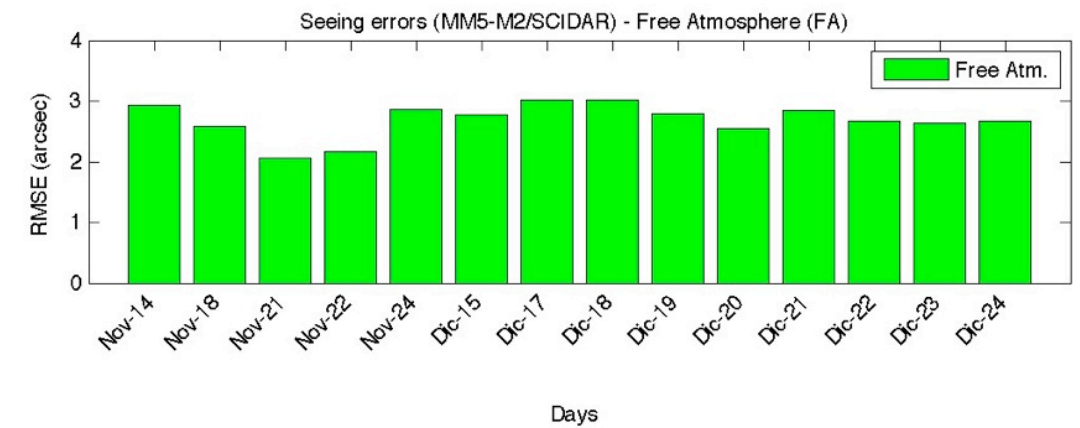
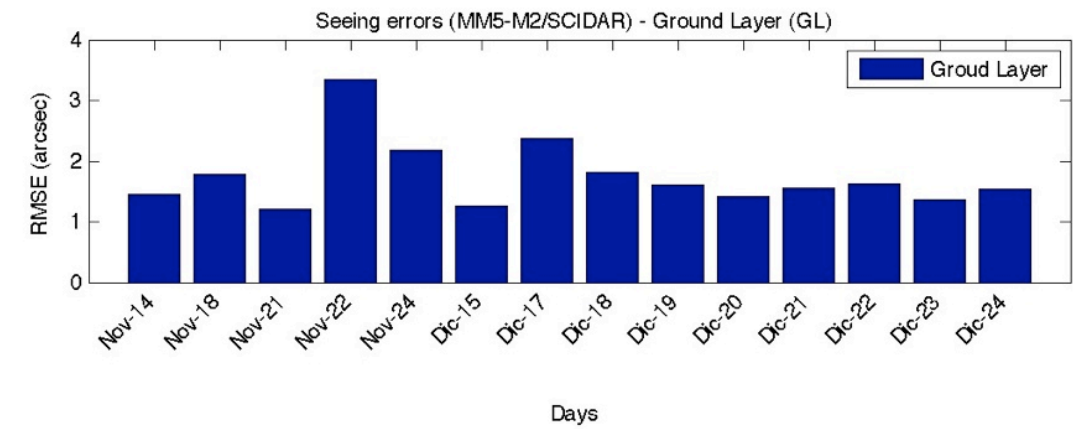
FA > 1 km

M1

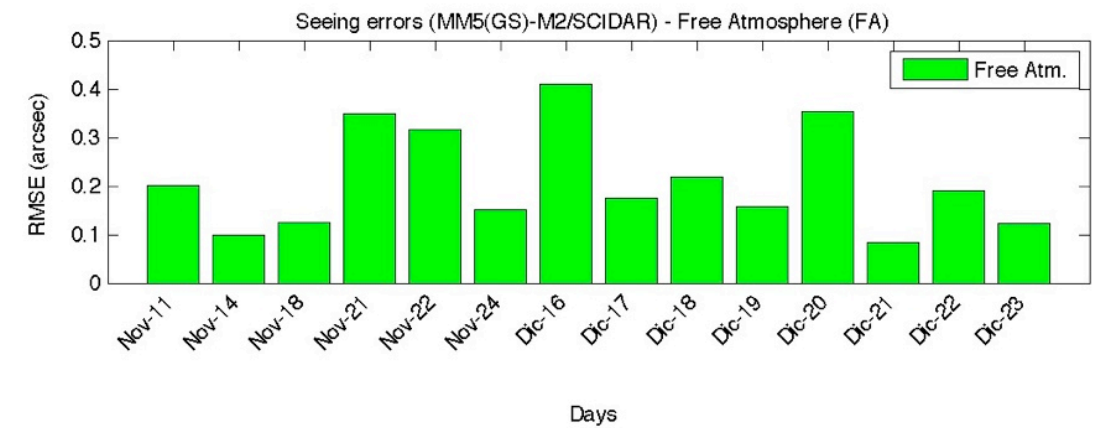
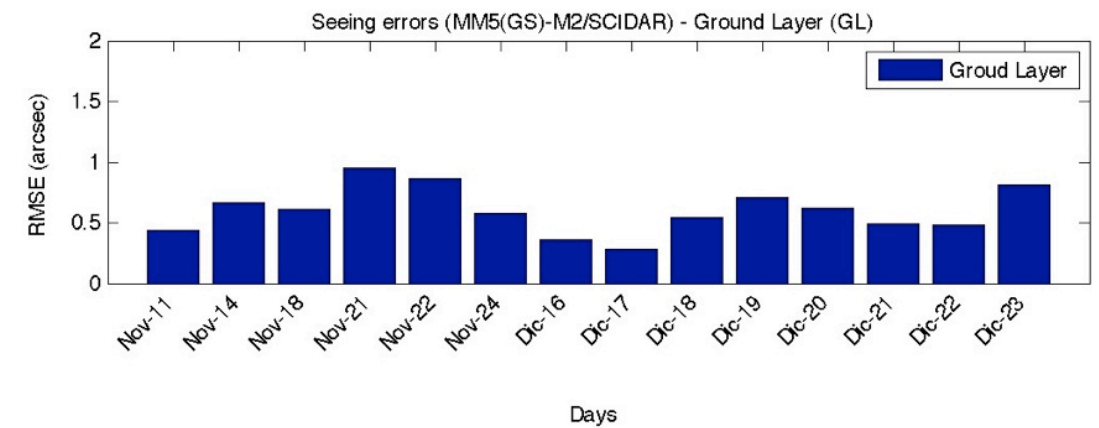
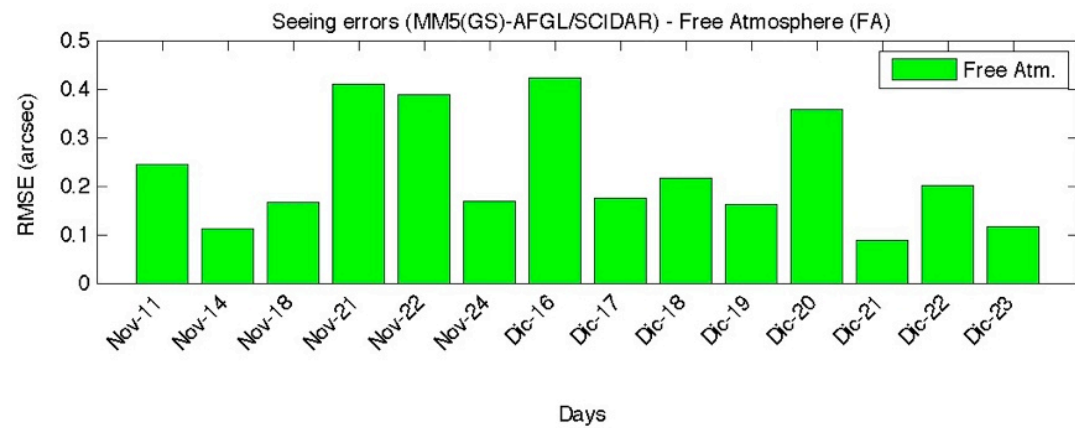
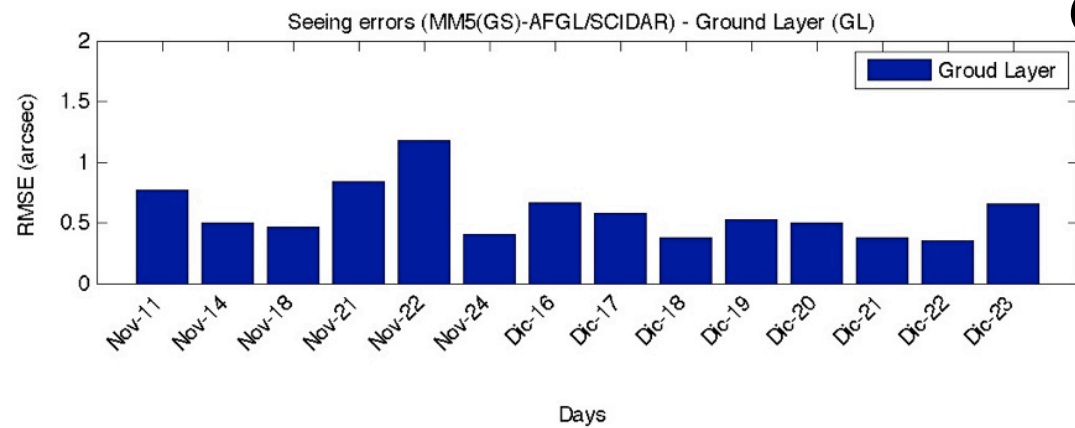


ETA

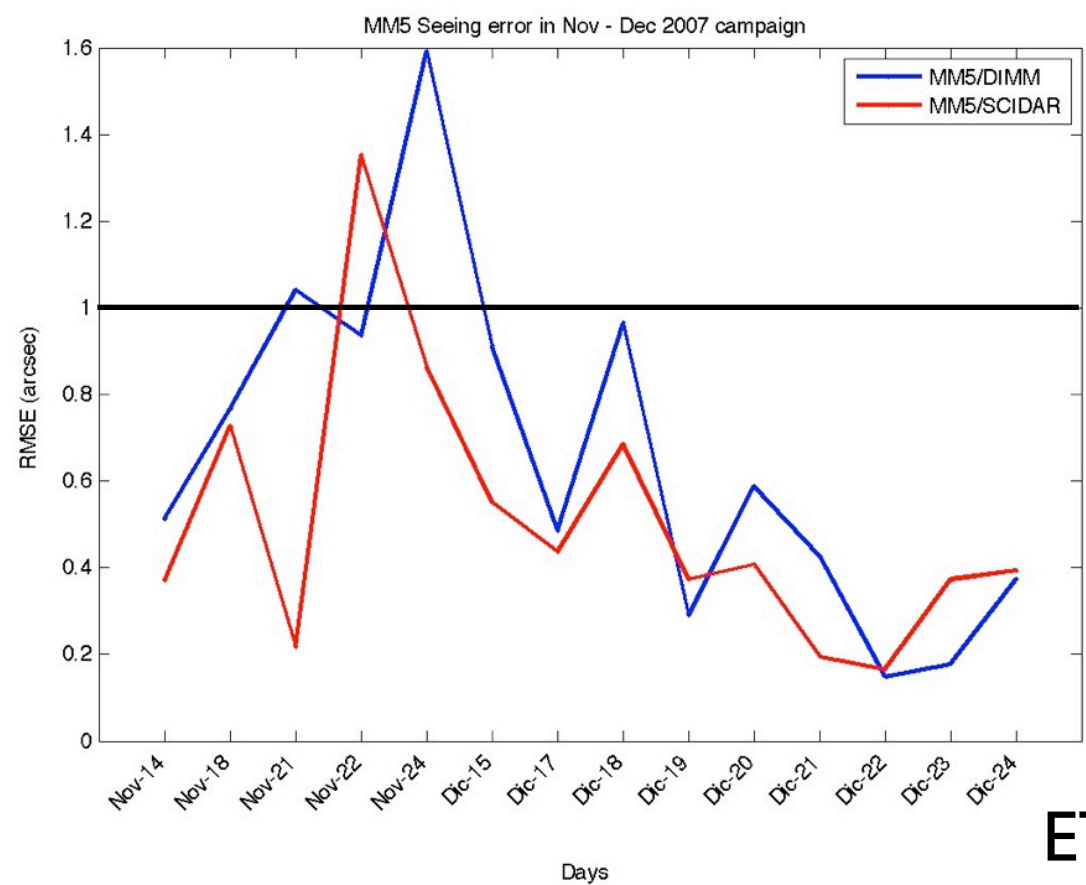
M2



GS

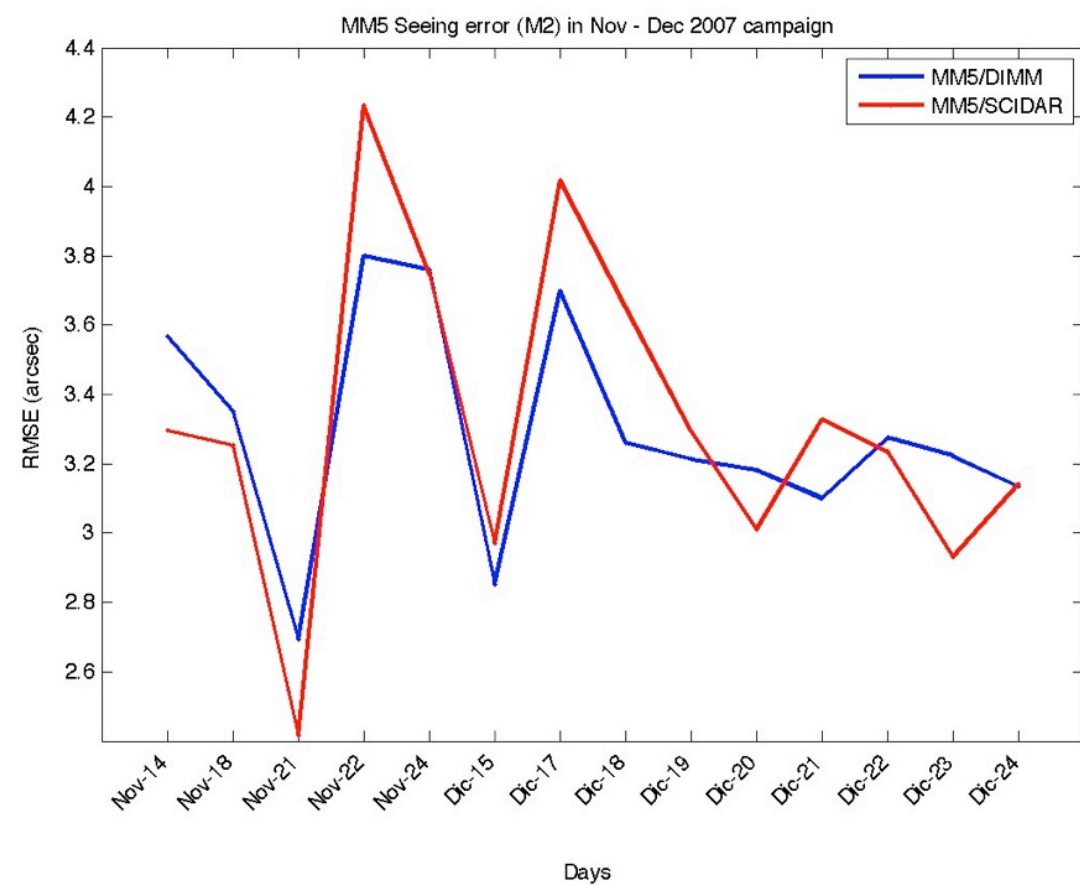


AFGL

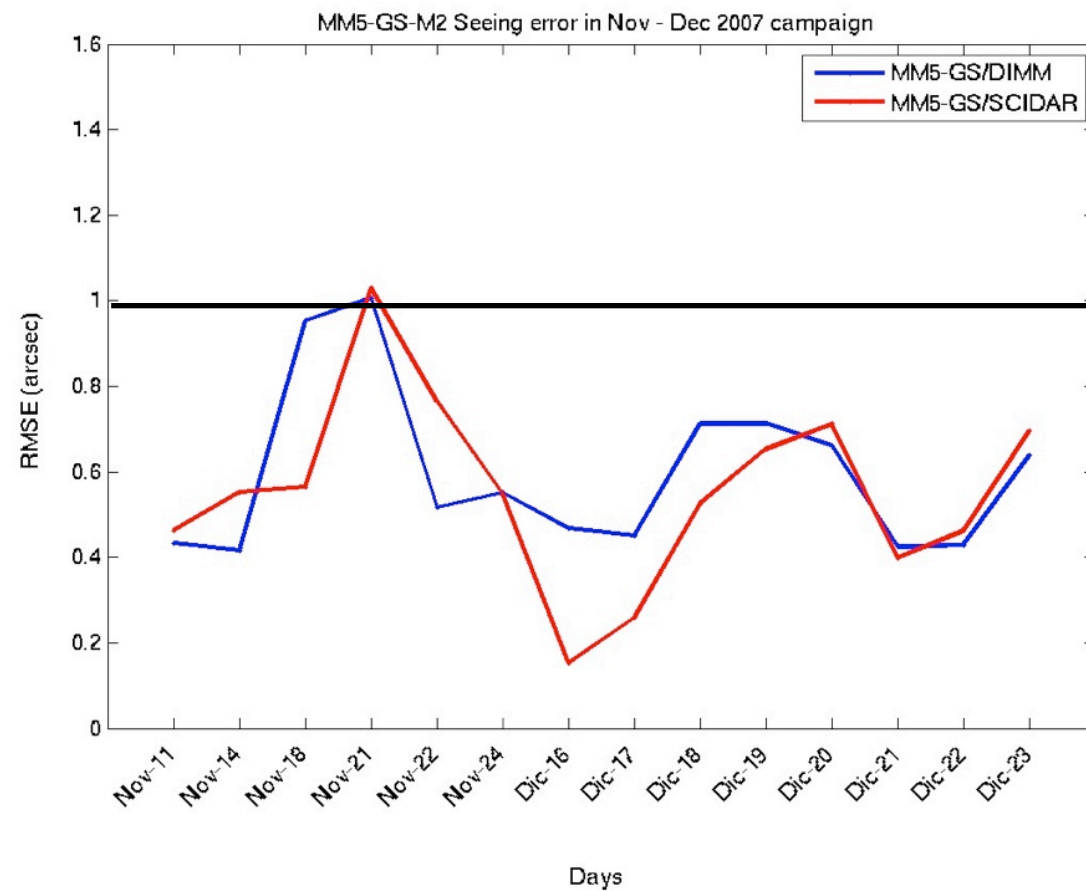
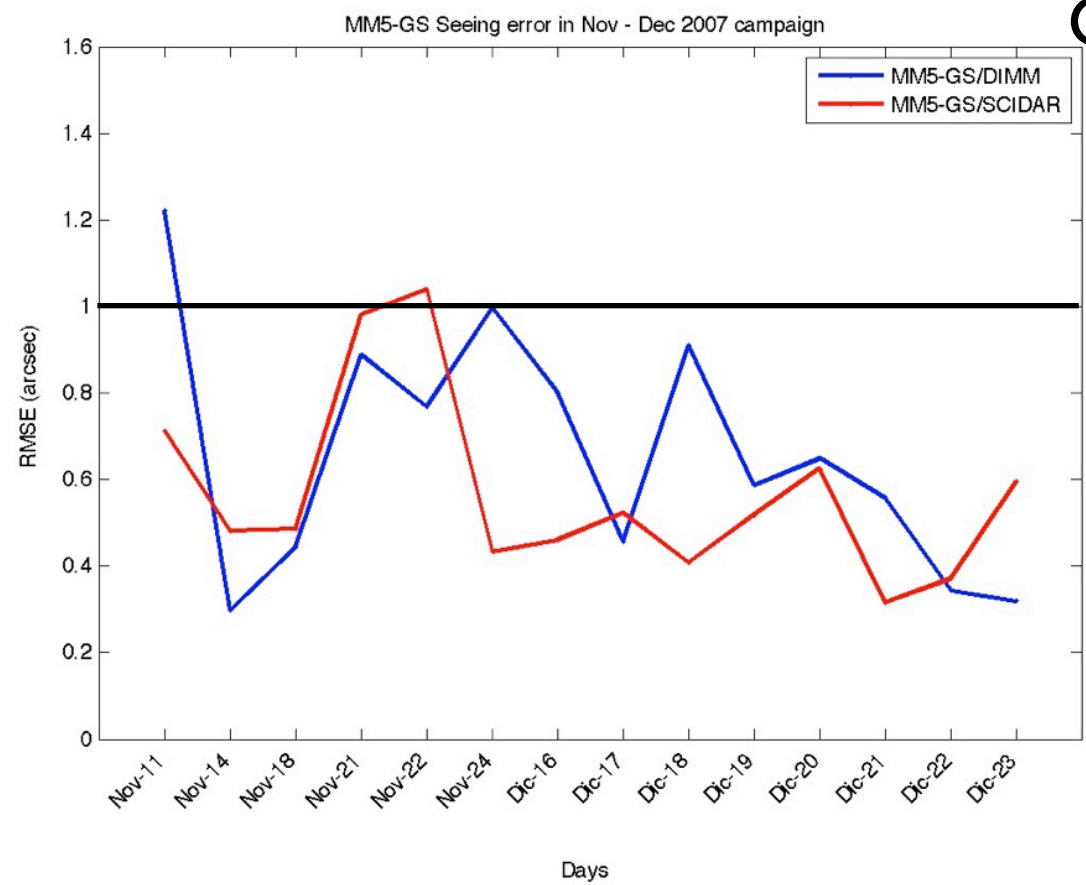


ETA

M2

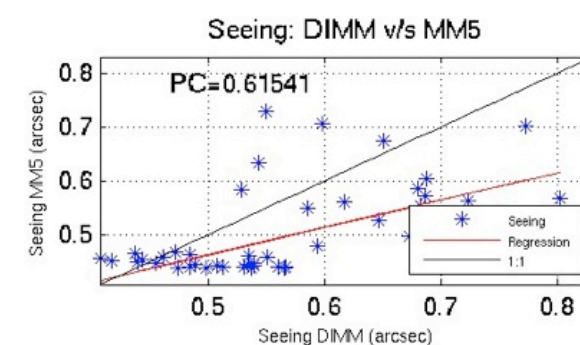
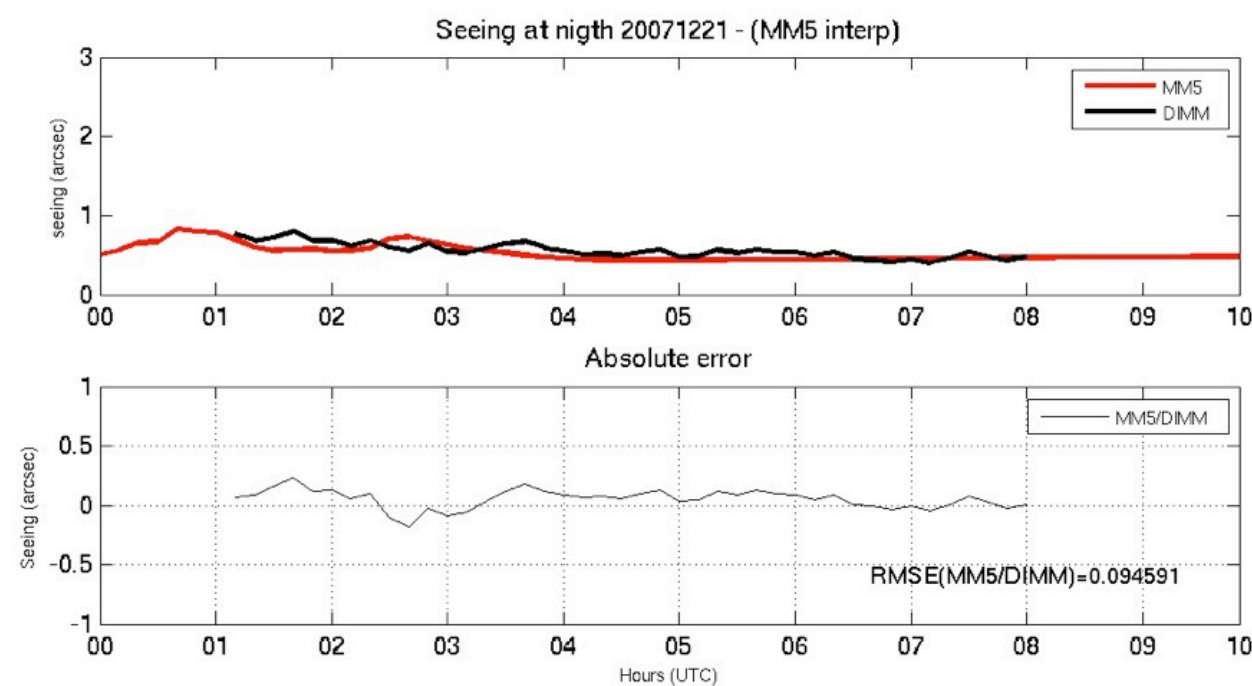
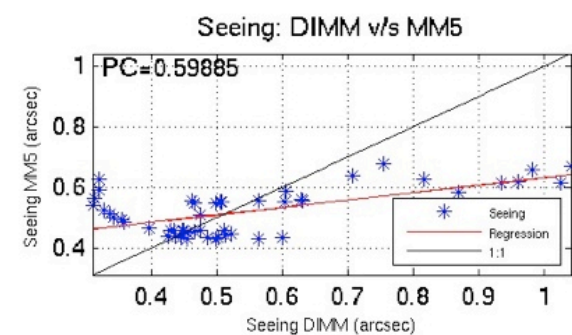
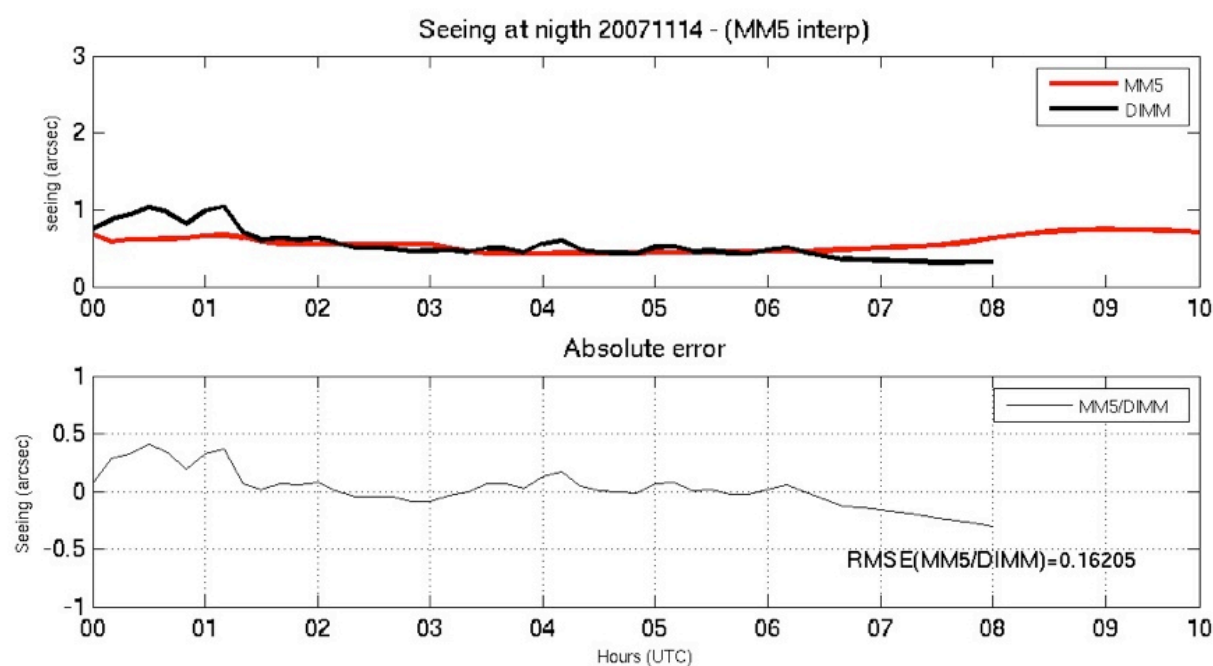


GS





Armazones M2 MM5/GS



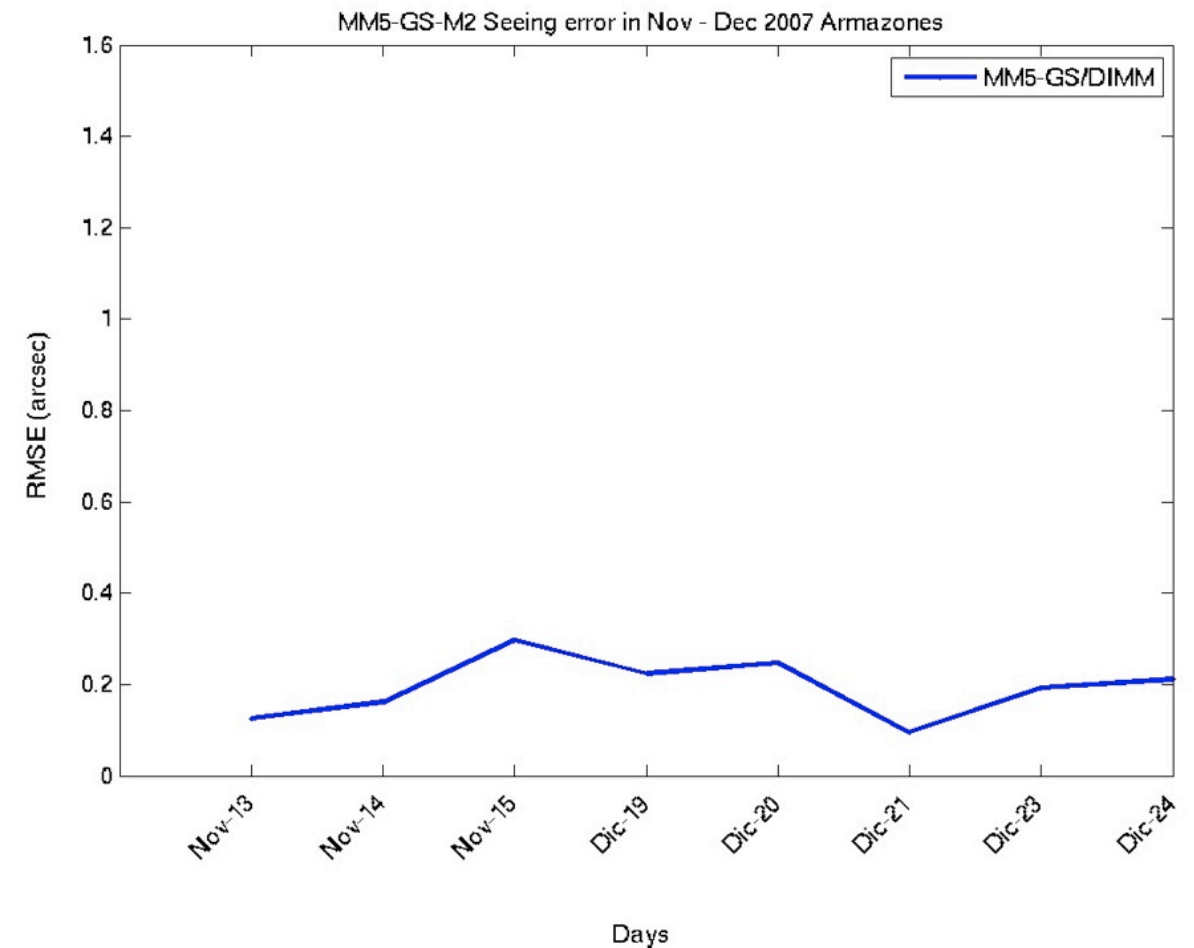
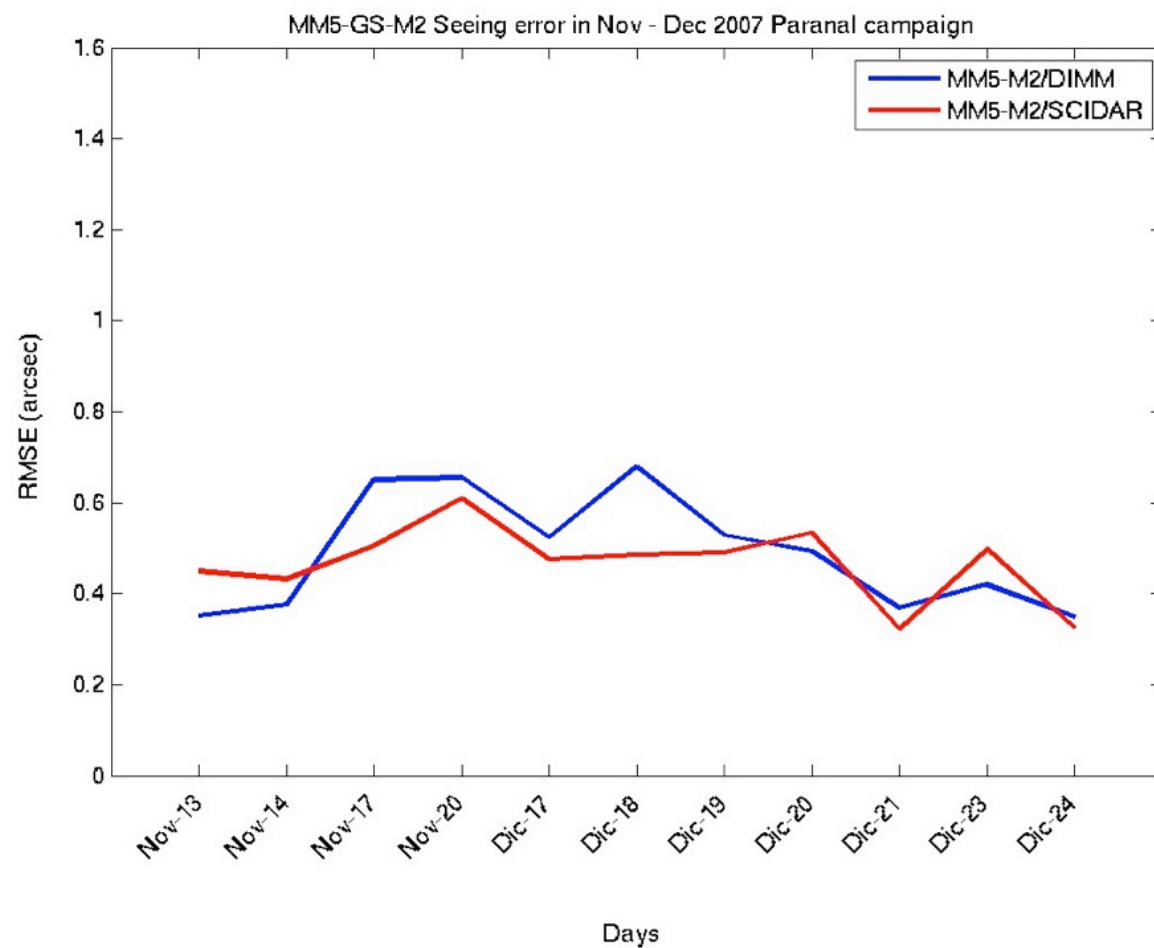


Next steps :

Testing other input data source
for SST and ALBEDO

Paranal RMSE $\sim 0.47''$

Armazones RMSE $\sim 0.2''$





Summary of results

1. M1 and M2 models have deficiencies in the Boundary Layer on Paranal, systematic error of the simulations is present.
2. The M2 with MM5/GS has smaller C_n^2 error in the free atmosphere, moreover it is better on jet stream zone.
3. Gayno-Seaman PBL is better than ETA PBL, specially in TKE for the C_n^2
4. The GL seeing shows more error than FA seeing.
5. M2/GS method shows the best seeing simulations, but it has big errors yet ($<1''$ in Paranal).
6. Using new source of input data (SST and Albedo) the error is low: Paranal $\sim 0.47''$ and Armazones $\sim 0.2''$.